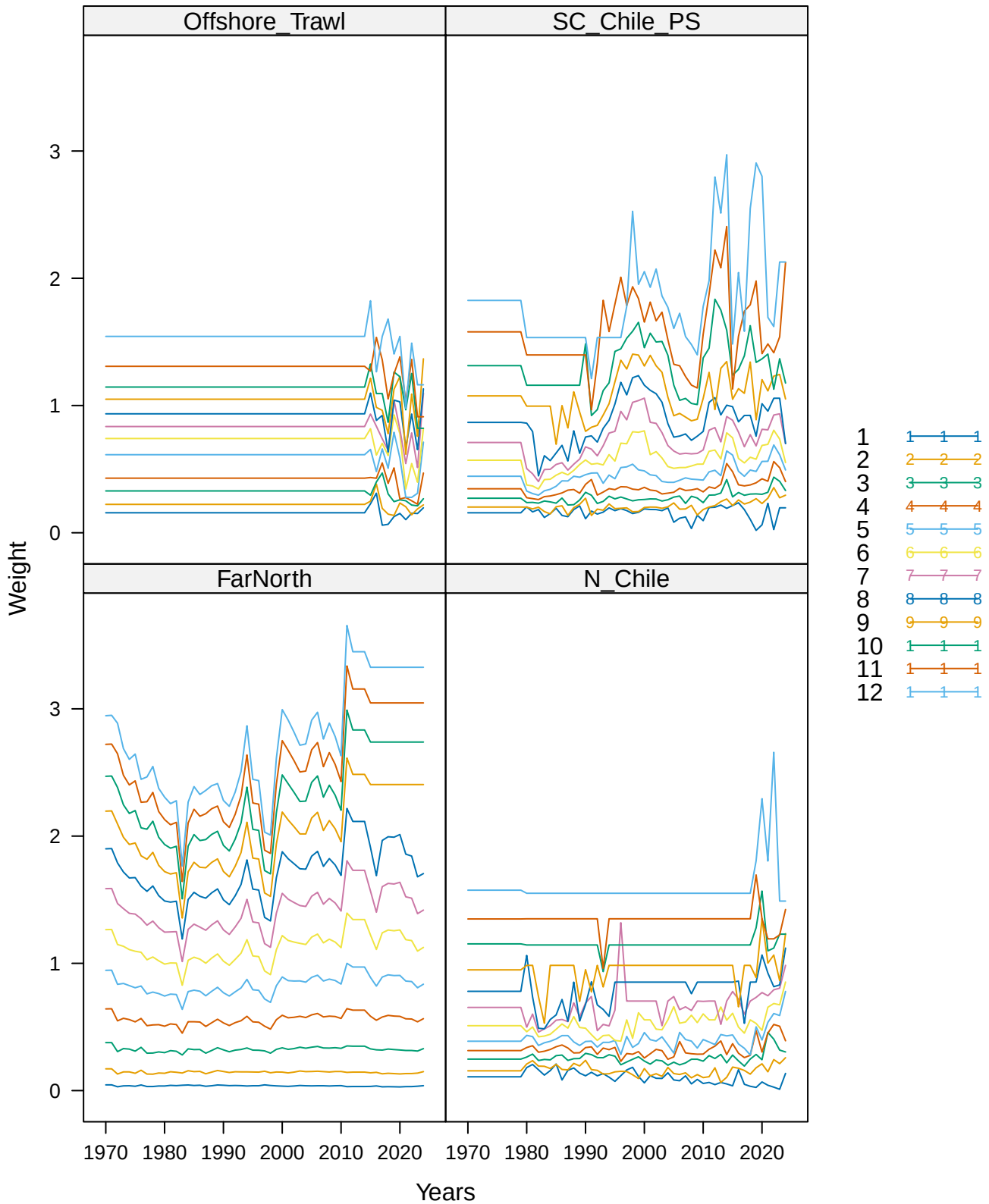
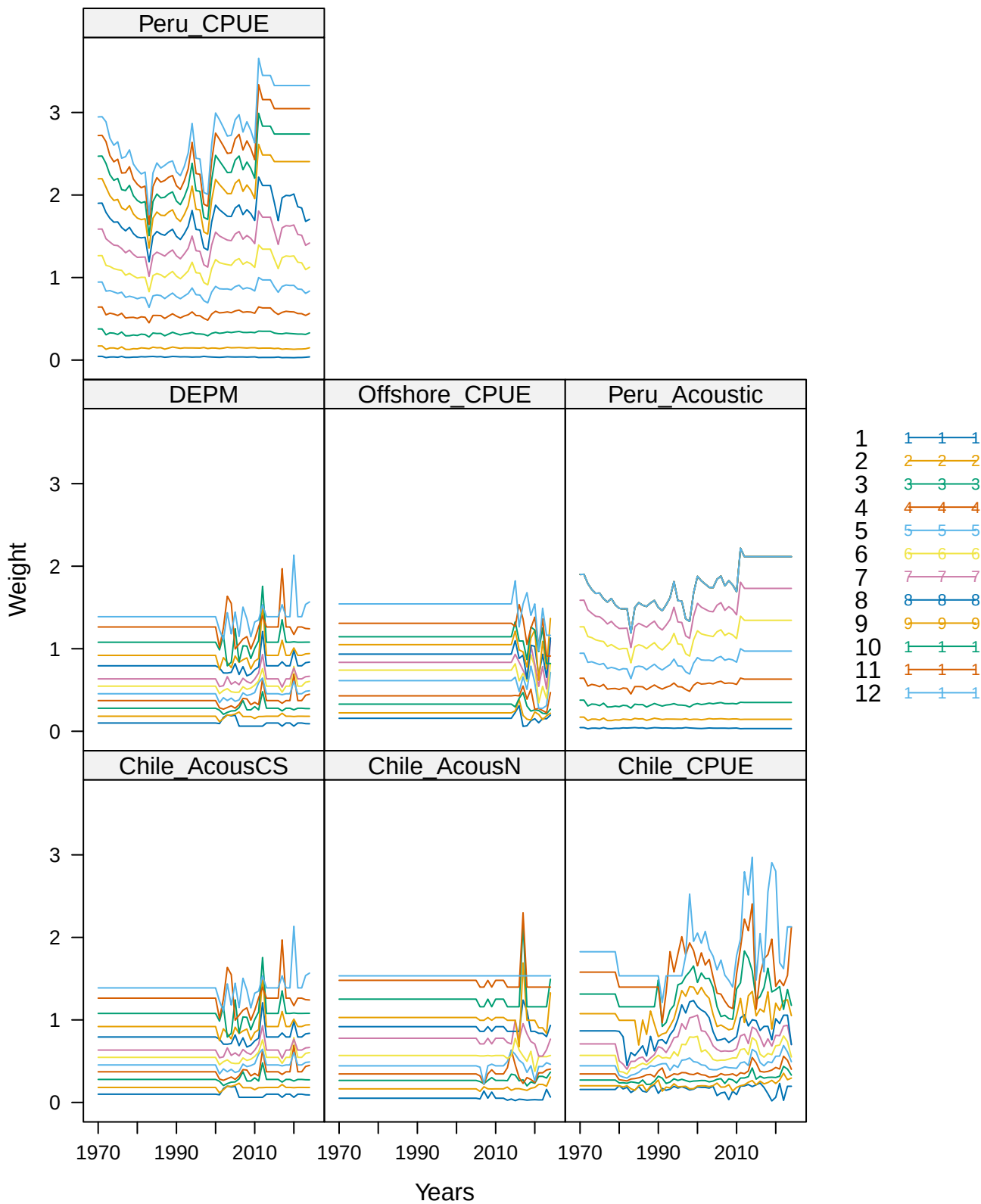


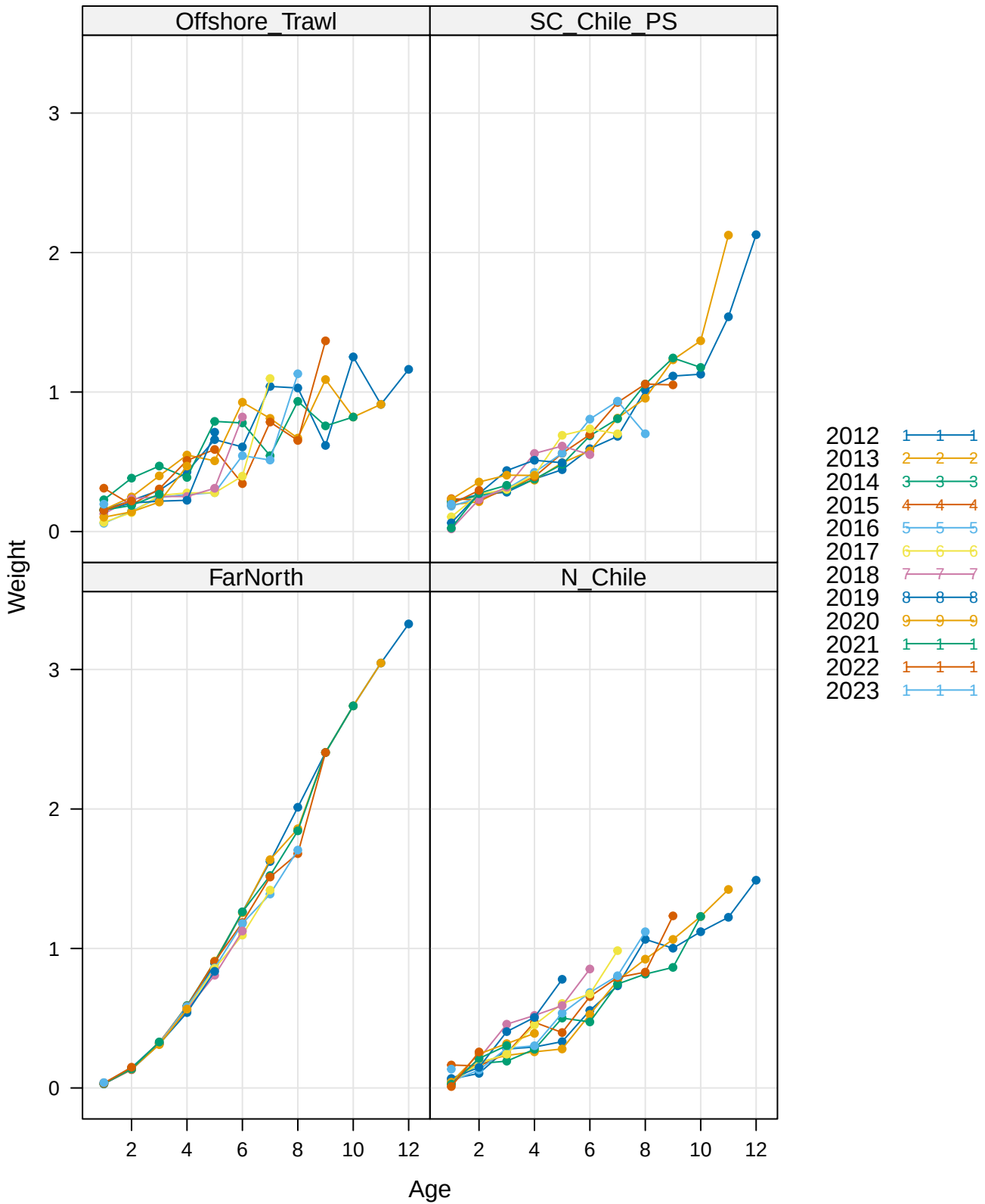
Weight at age in the fishery



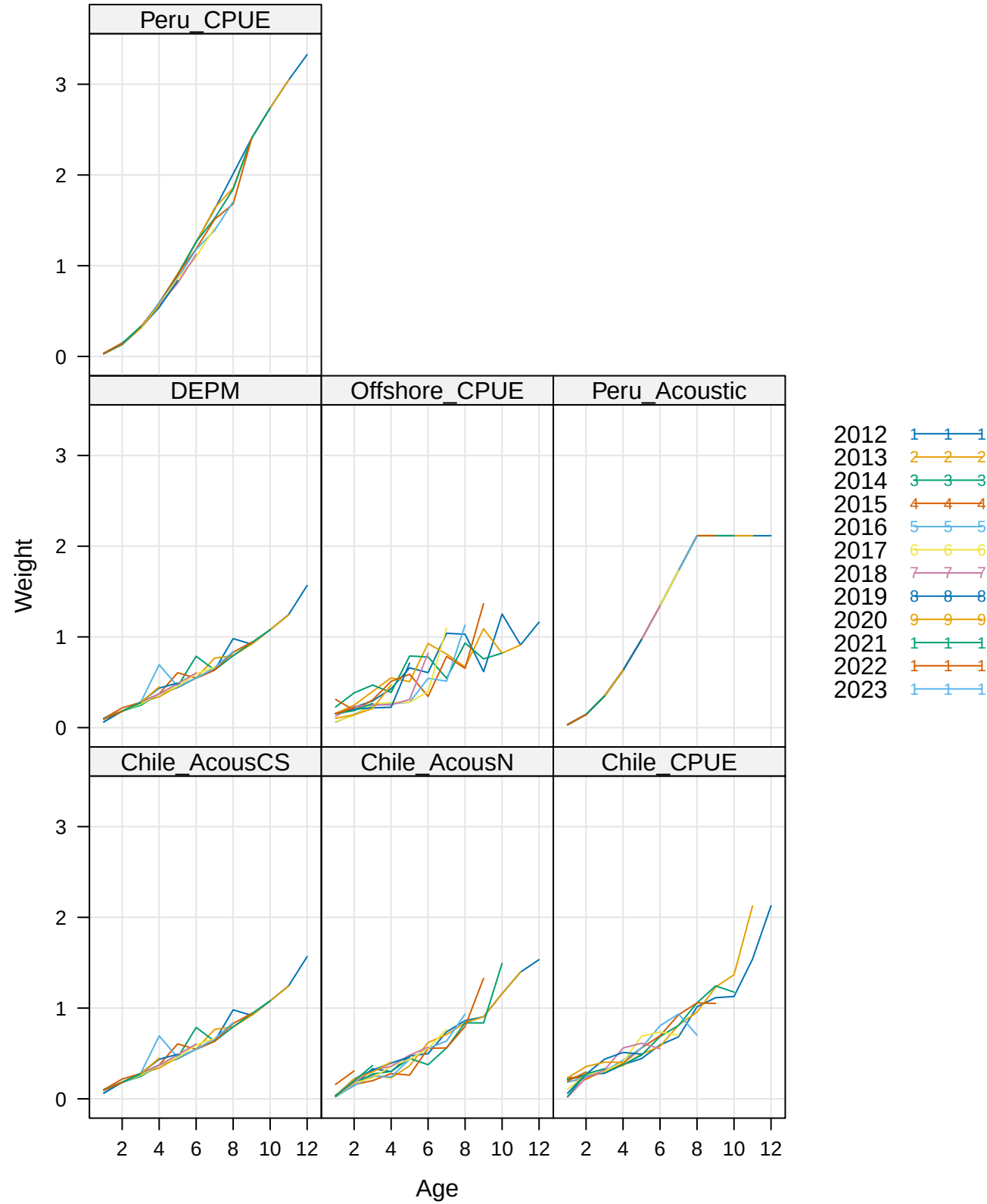
Weight at age in the survey



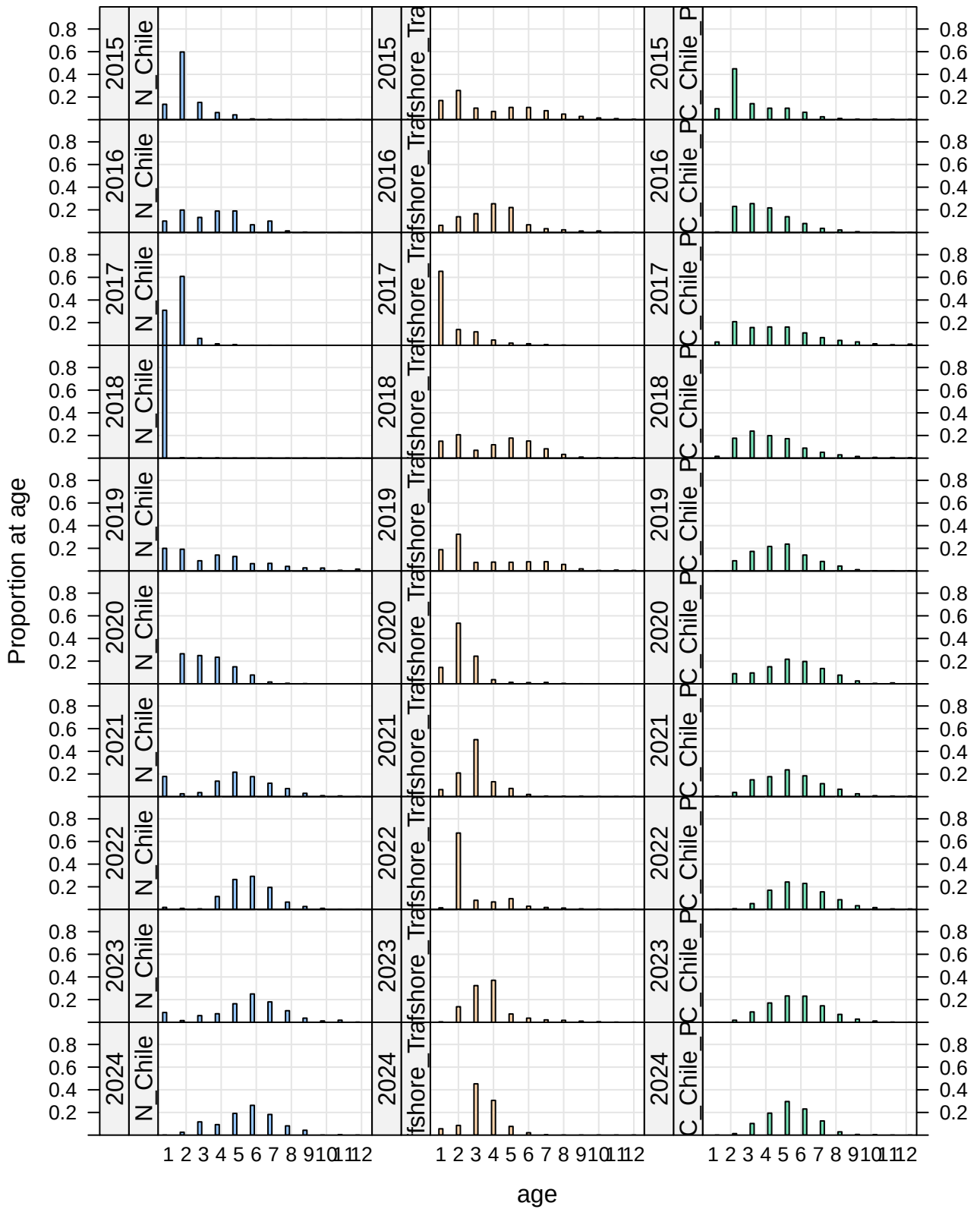
Weight at age by cohort in the fleet



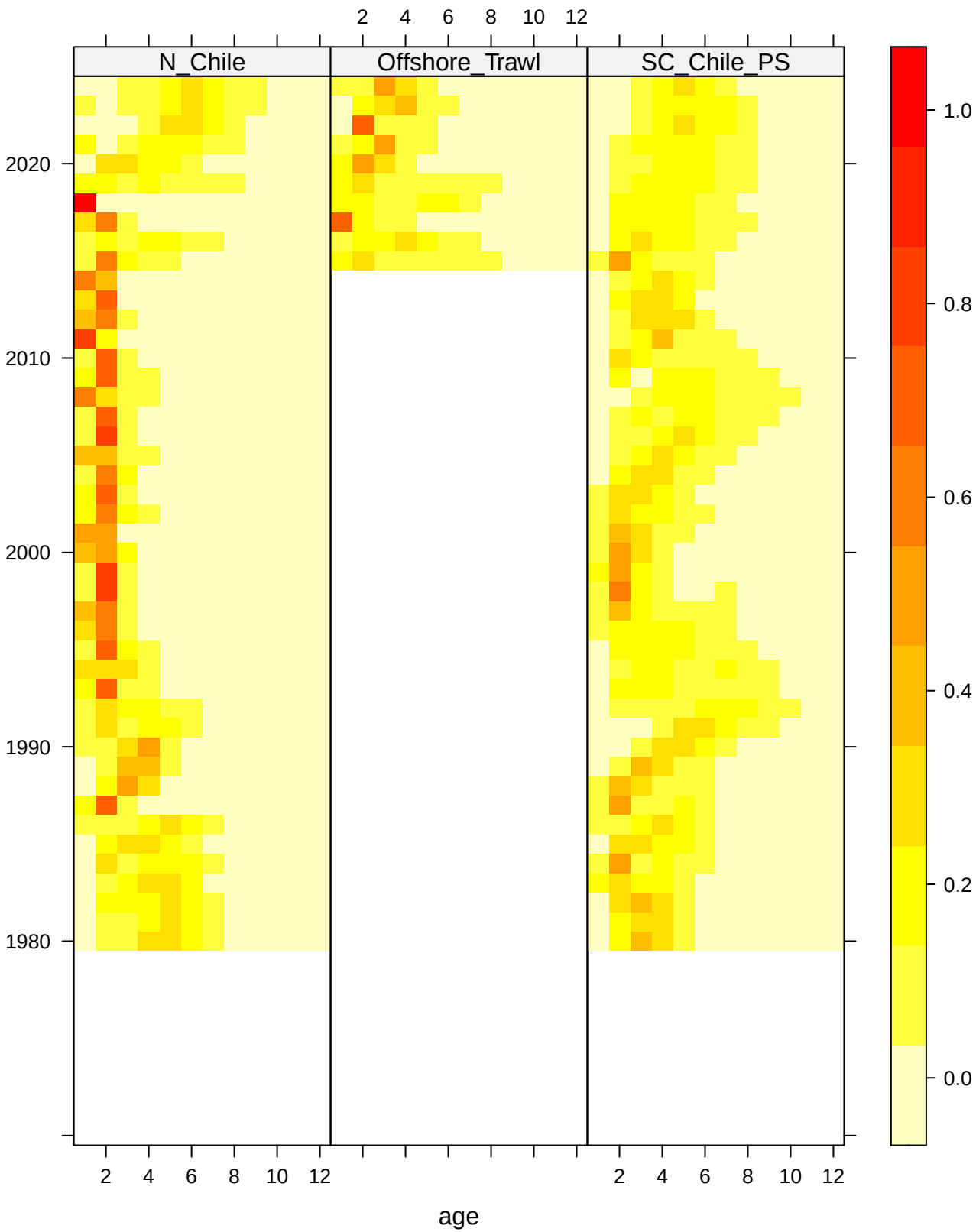
Weight at age by cohort in the survey



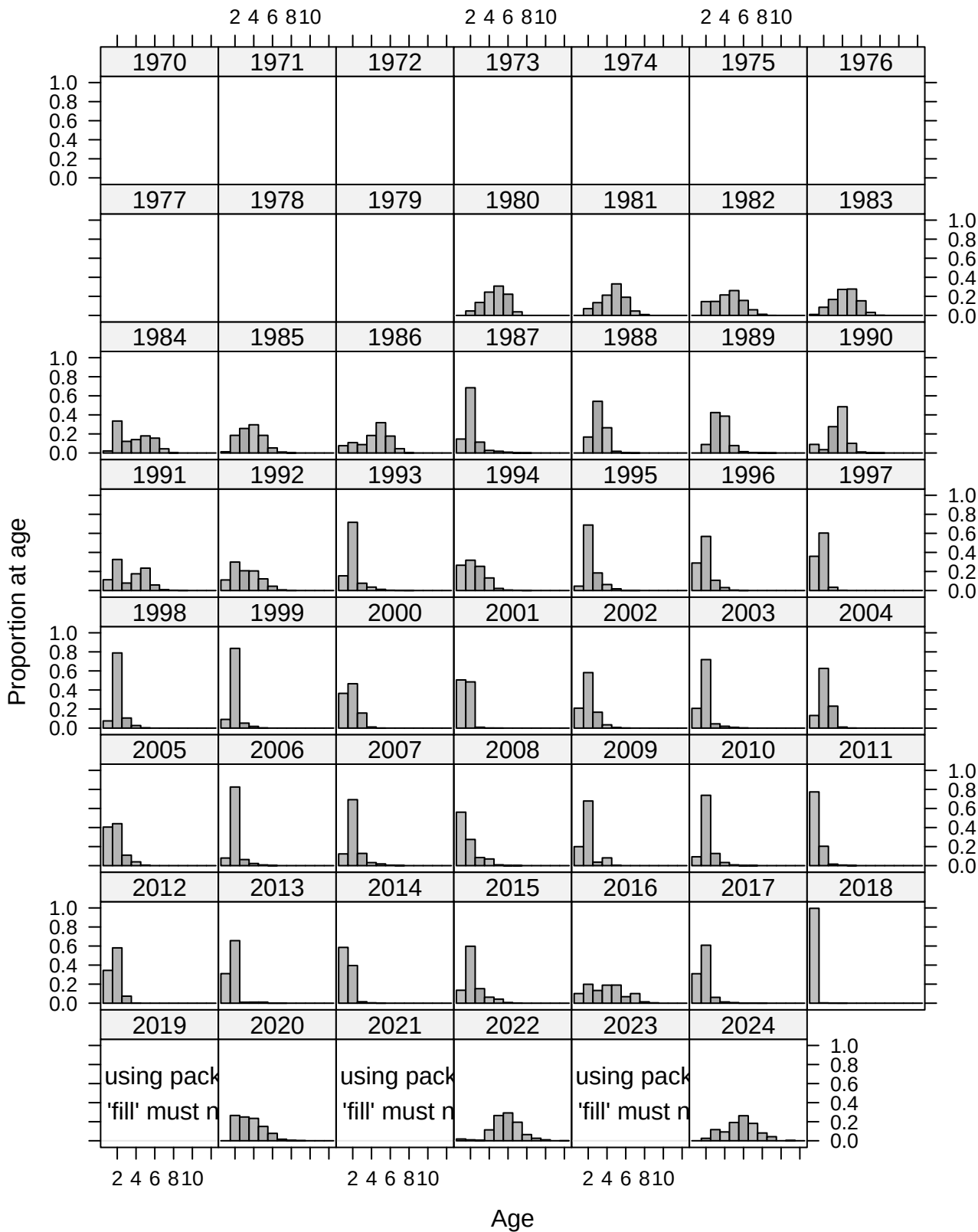
Age composition in fleets



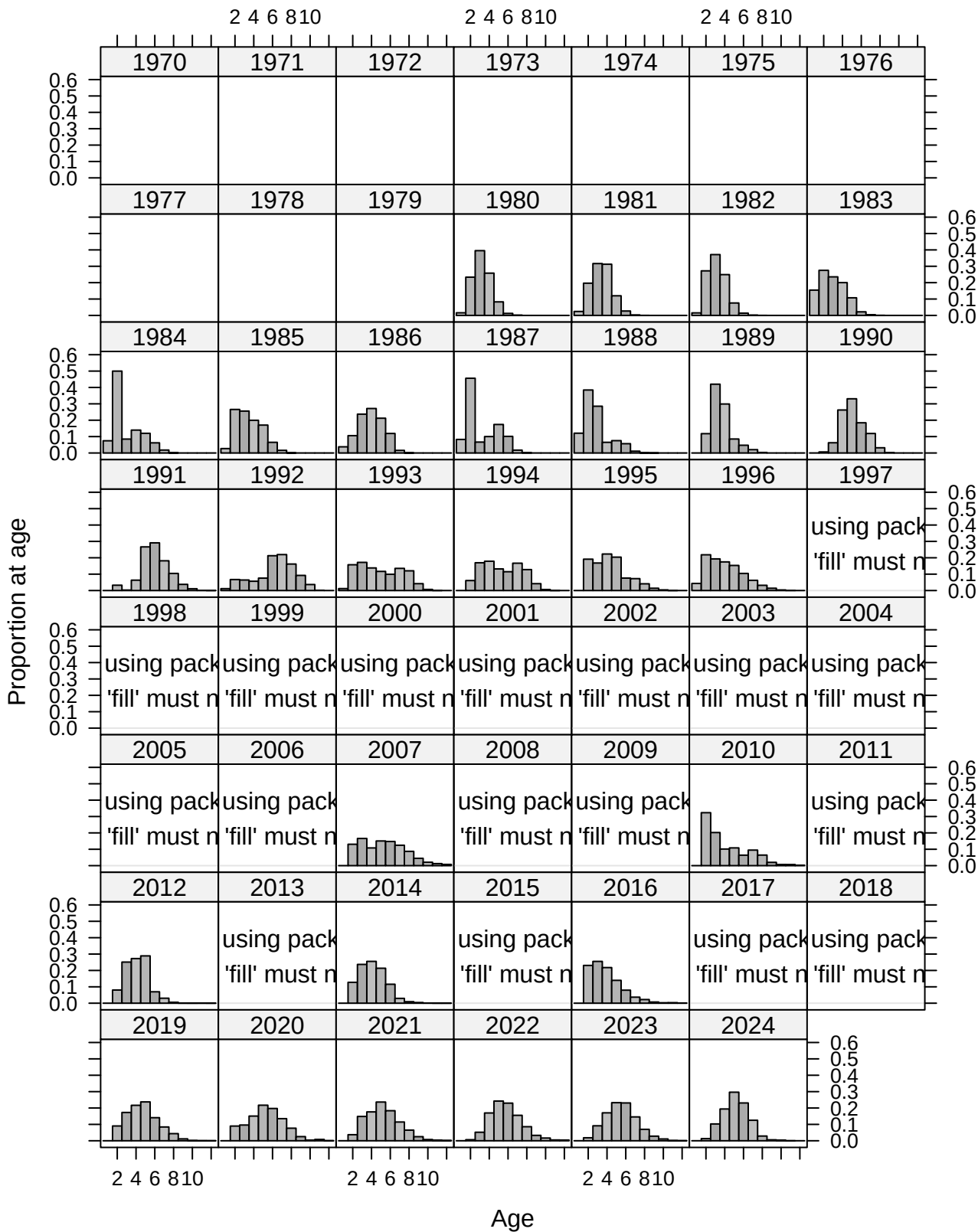
Age composition in fleets



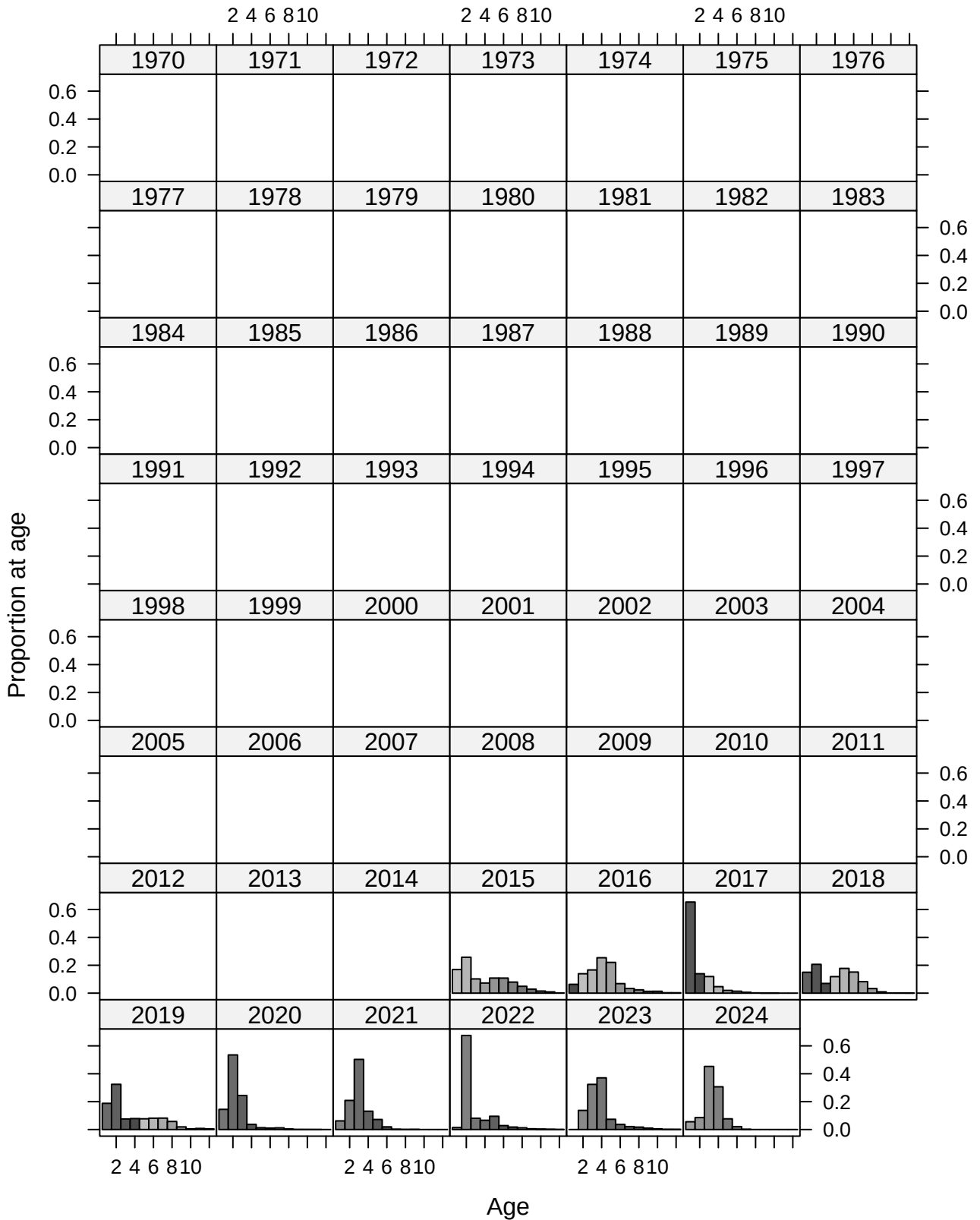
Age composition in fleets N_Chile



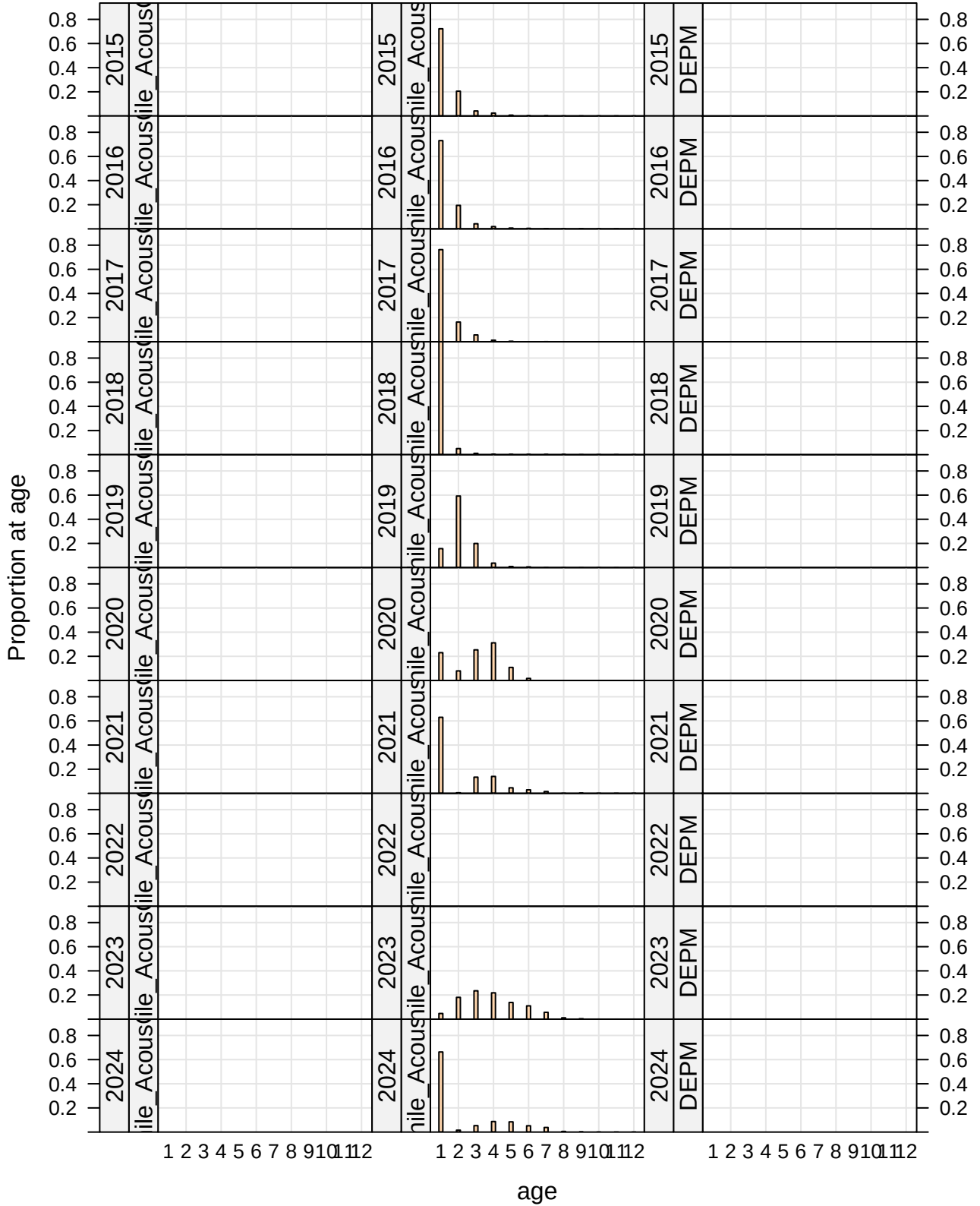
Age composition in fleets SC_Chile_PS



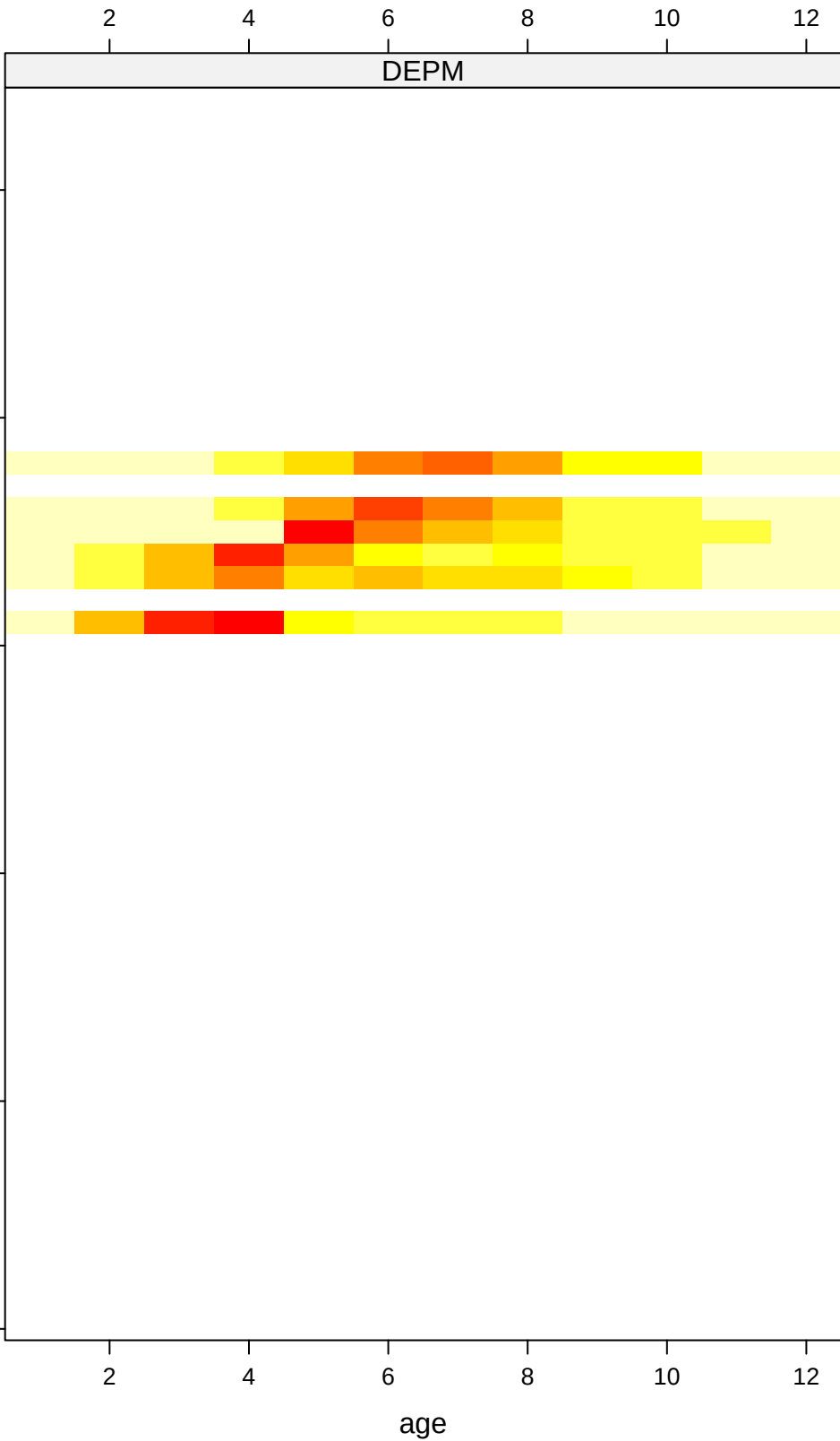
Age composition in fleets Offshore_Trawl



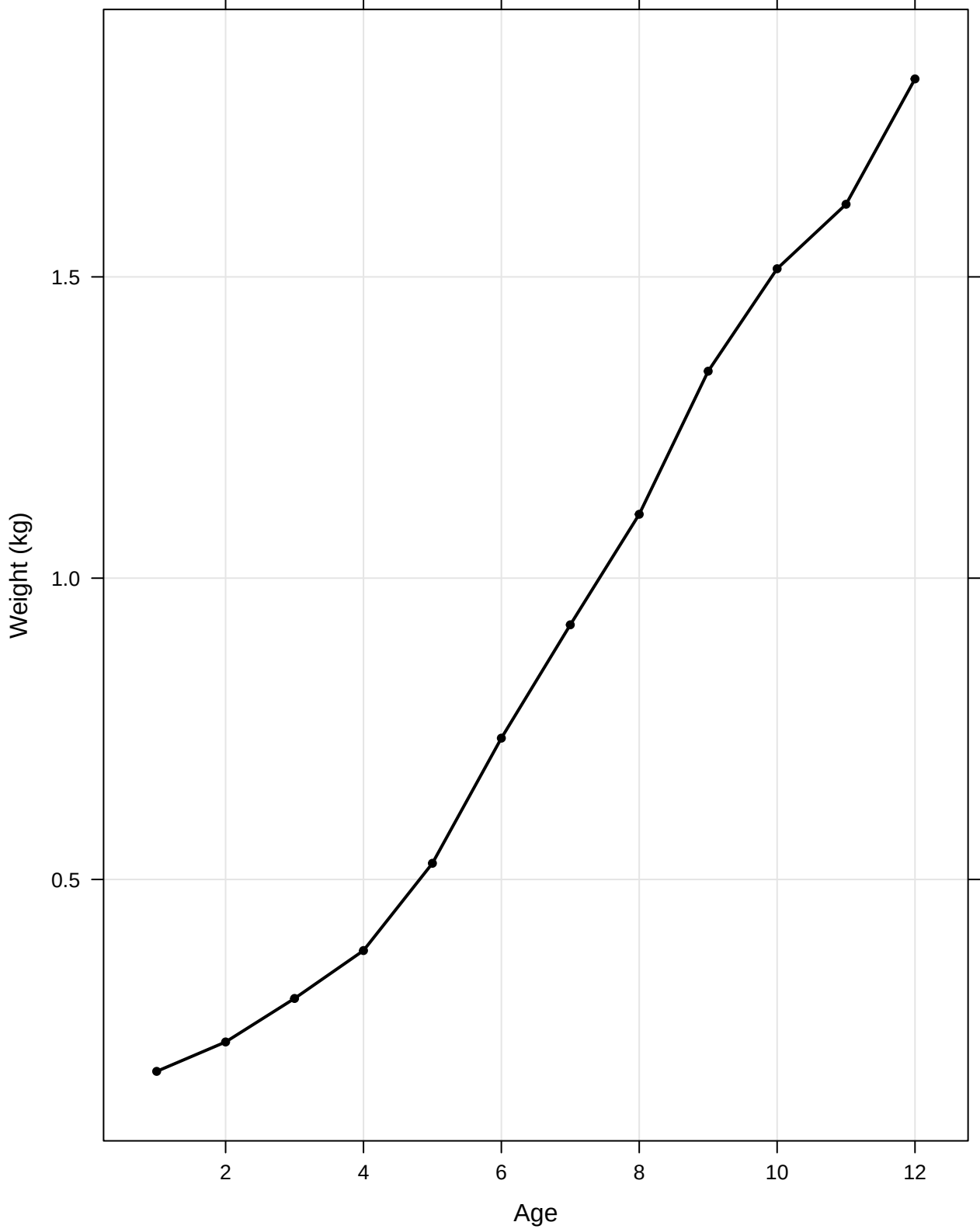
Age composition in surveys



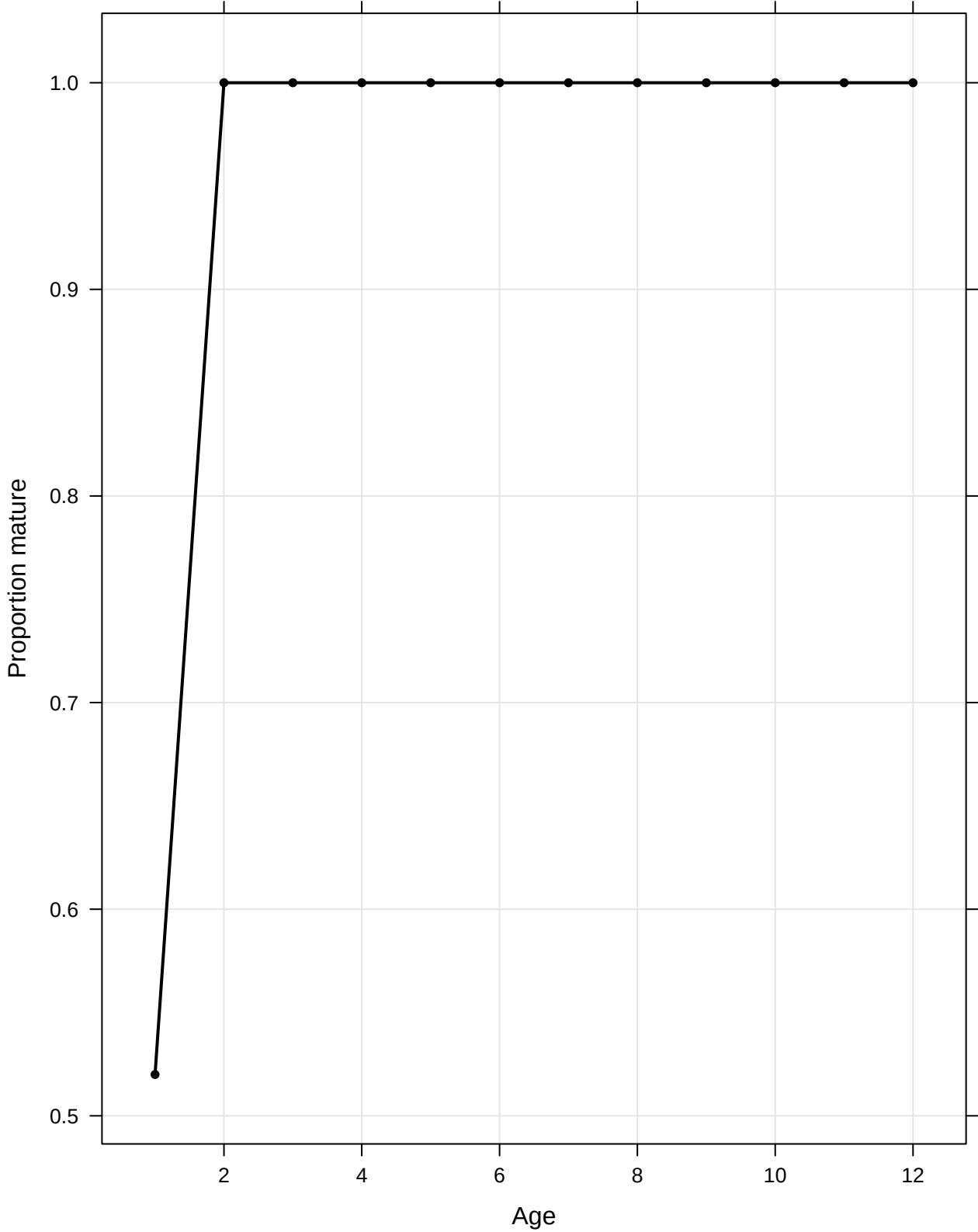
2 4 6 8 10 12



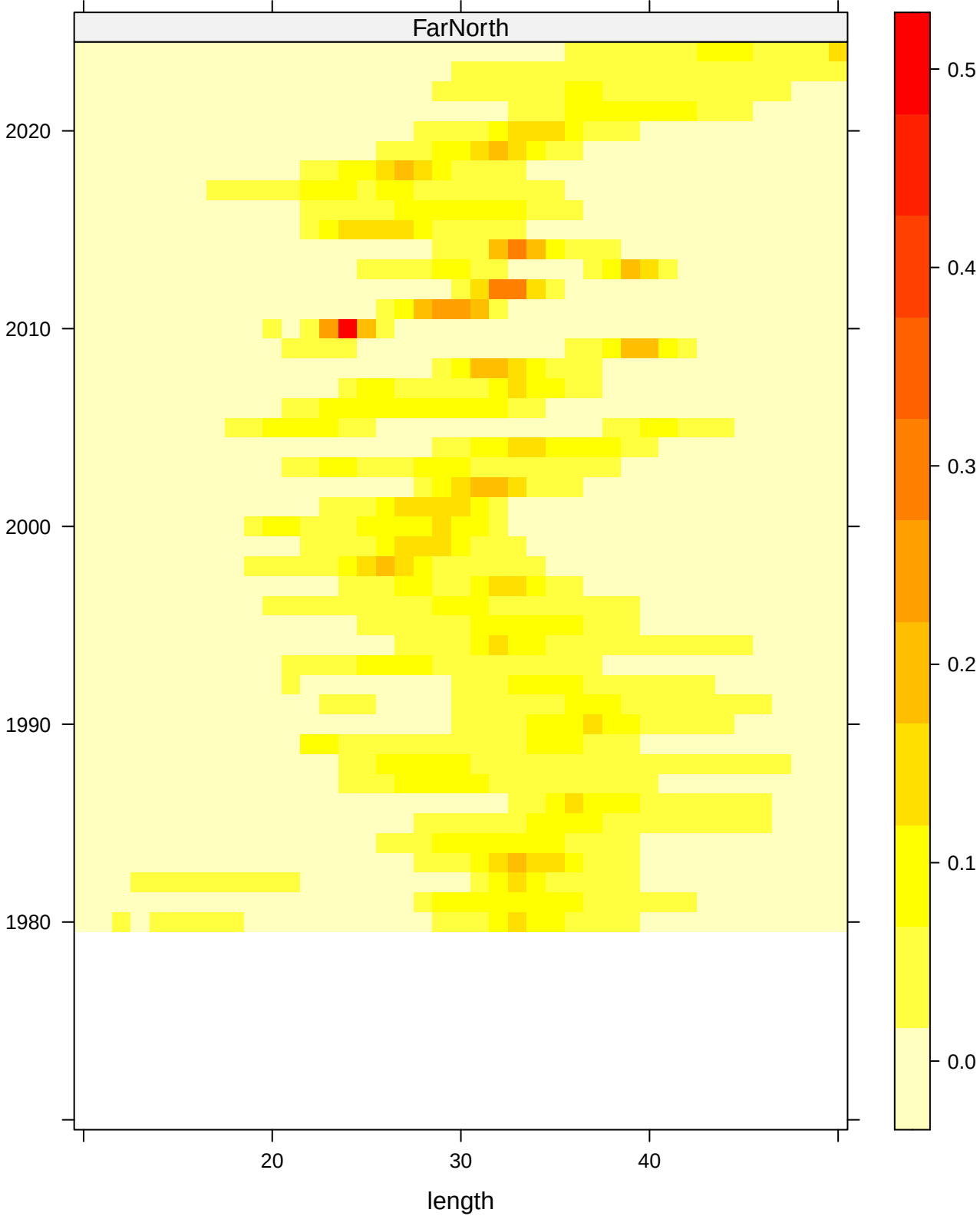
Weight in the stock



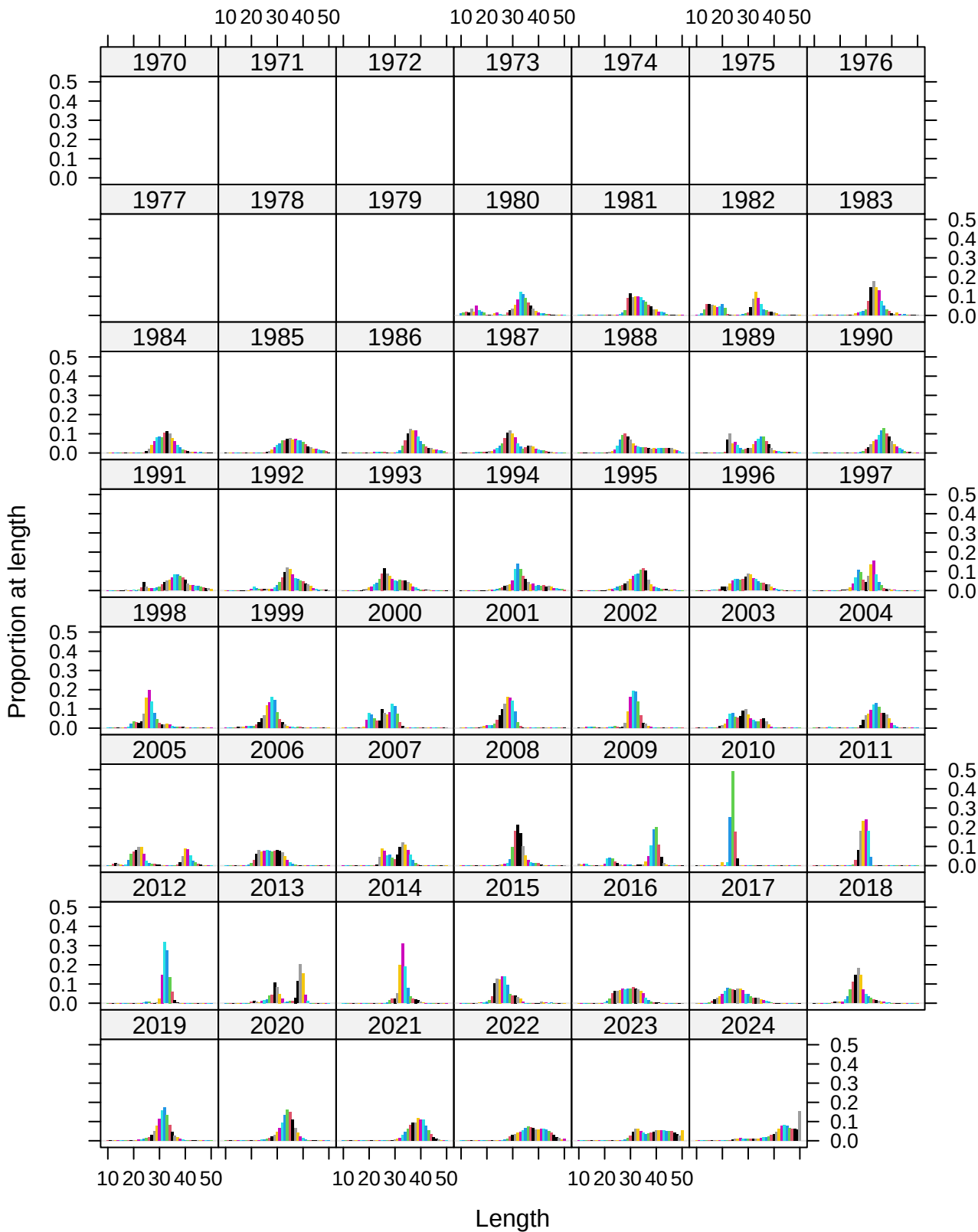
Maturity in the stock



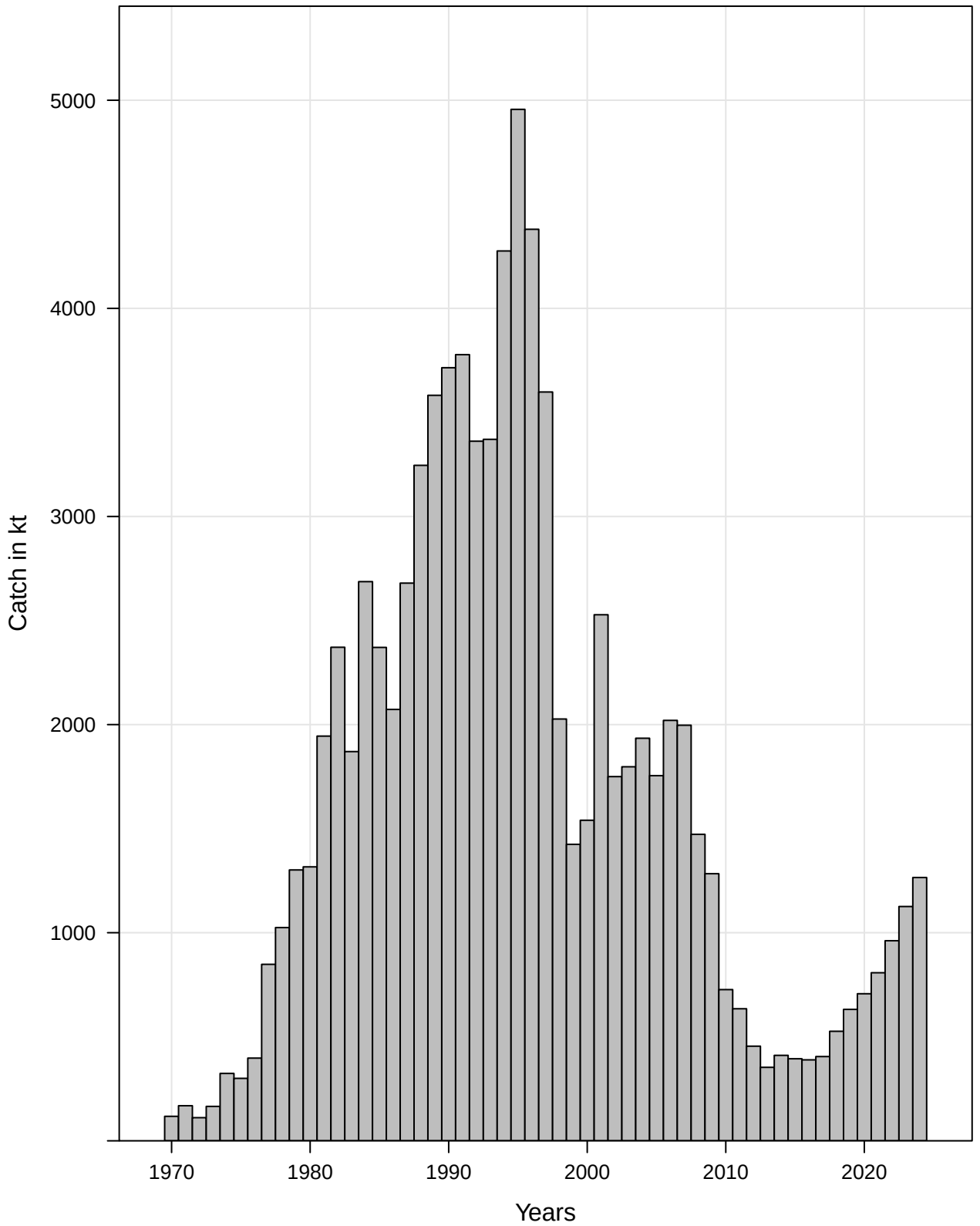
Length composition in fleet FarNorth



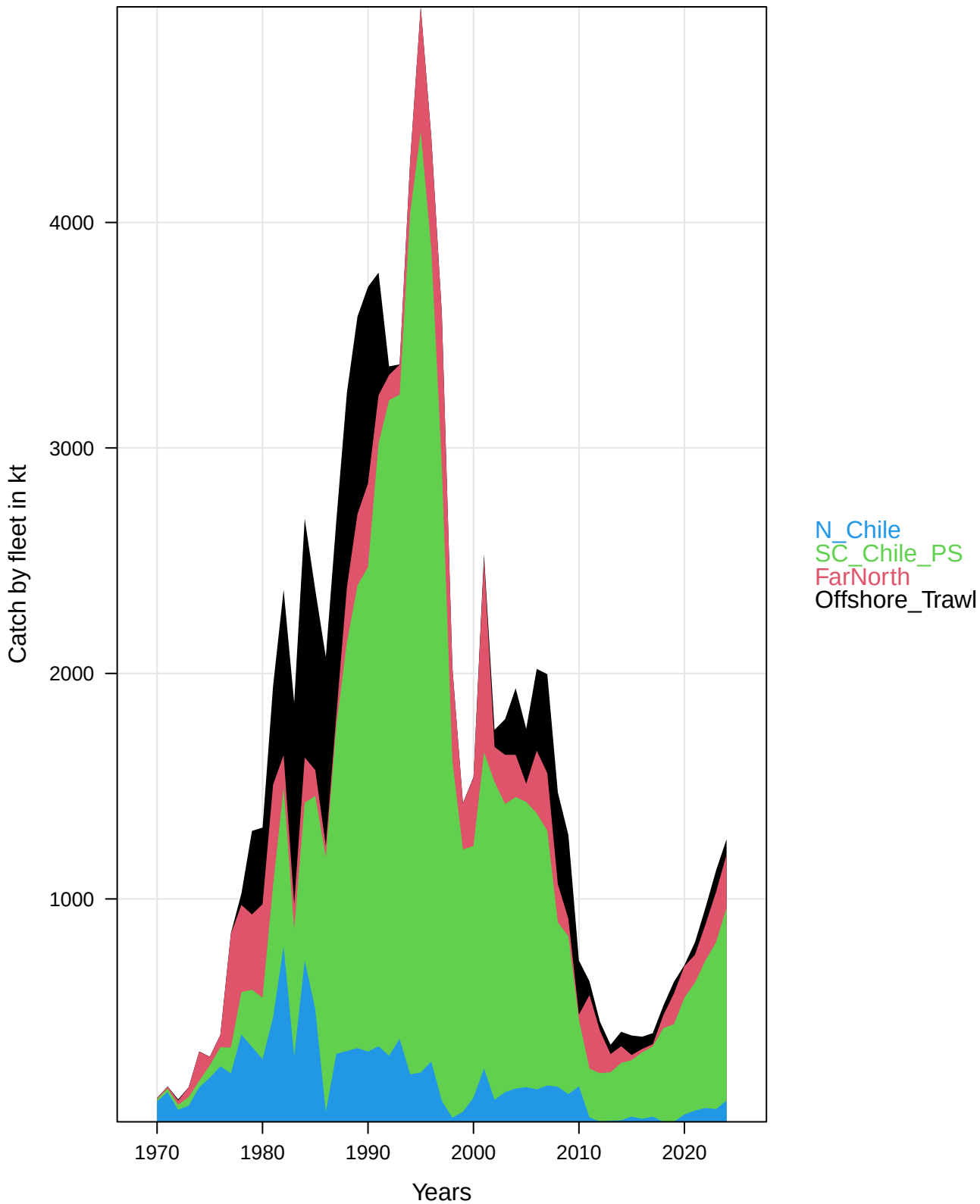
Length composition in fleets FarNorth



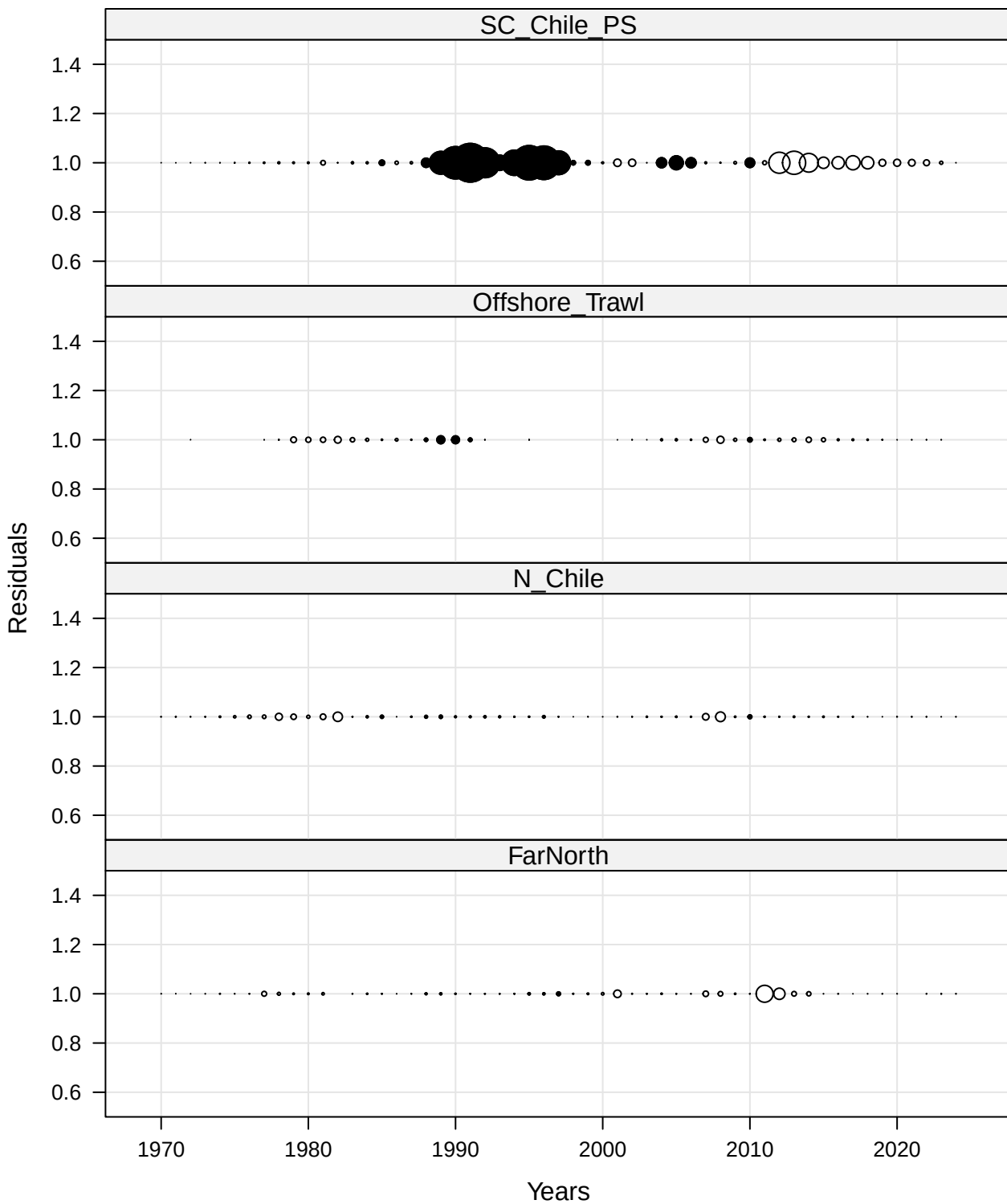
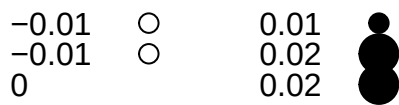
Total catch

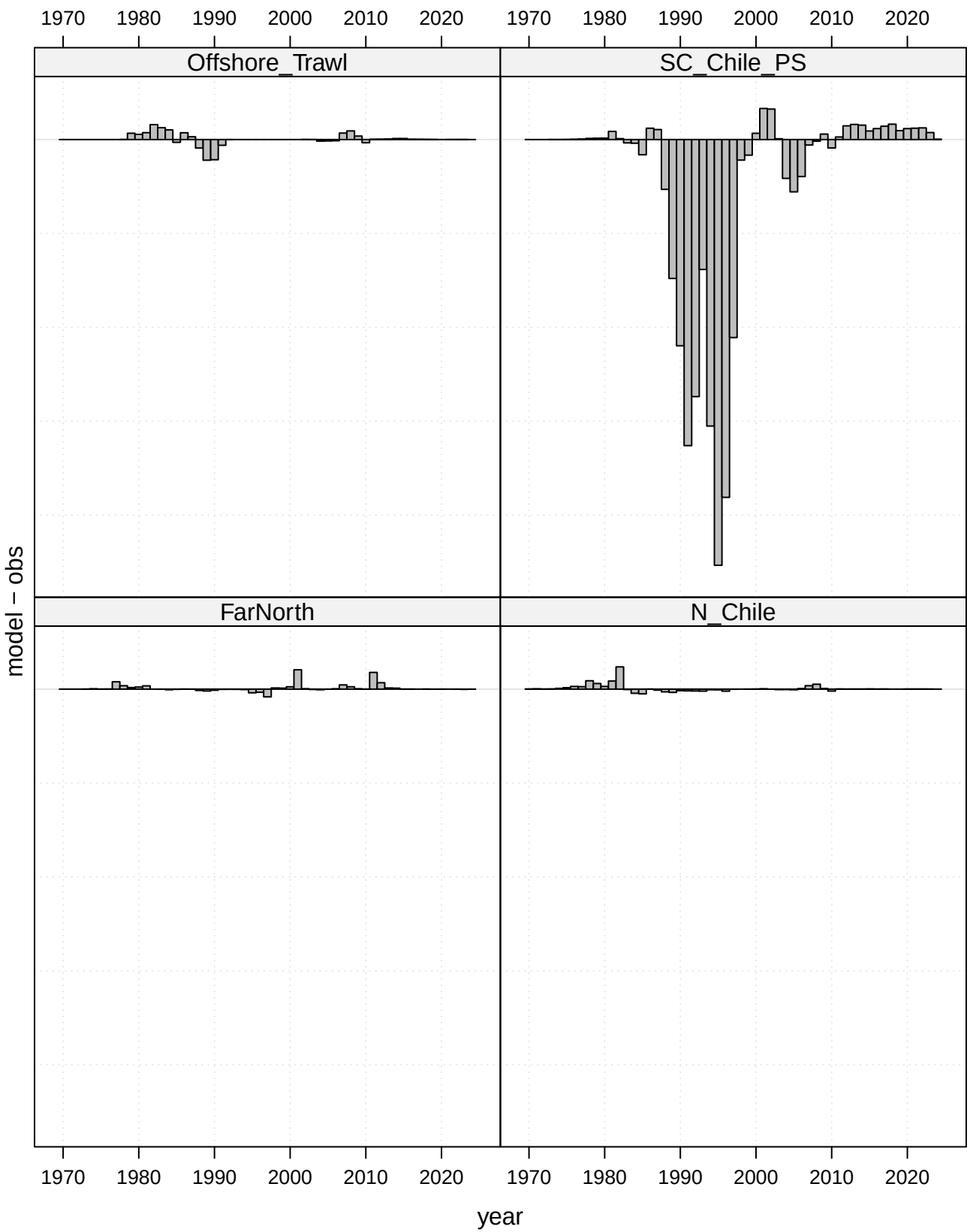


Total catch by fleet

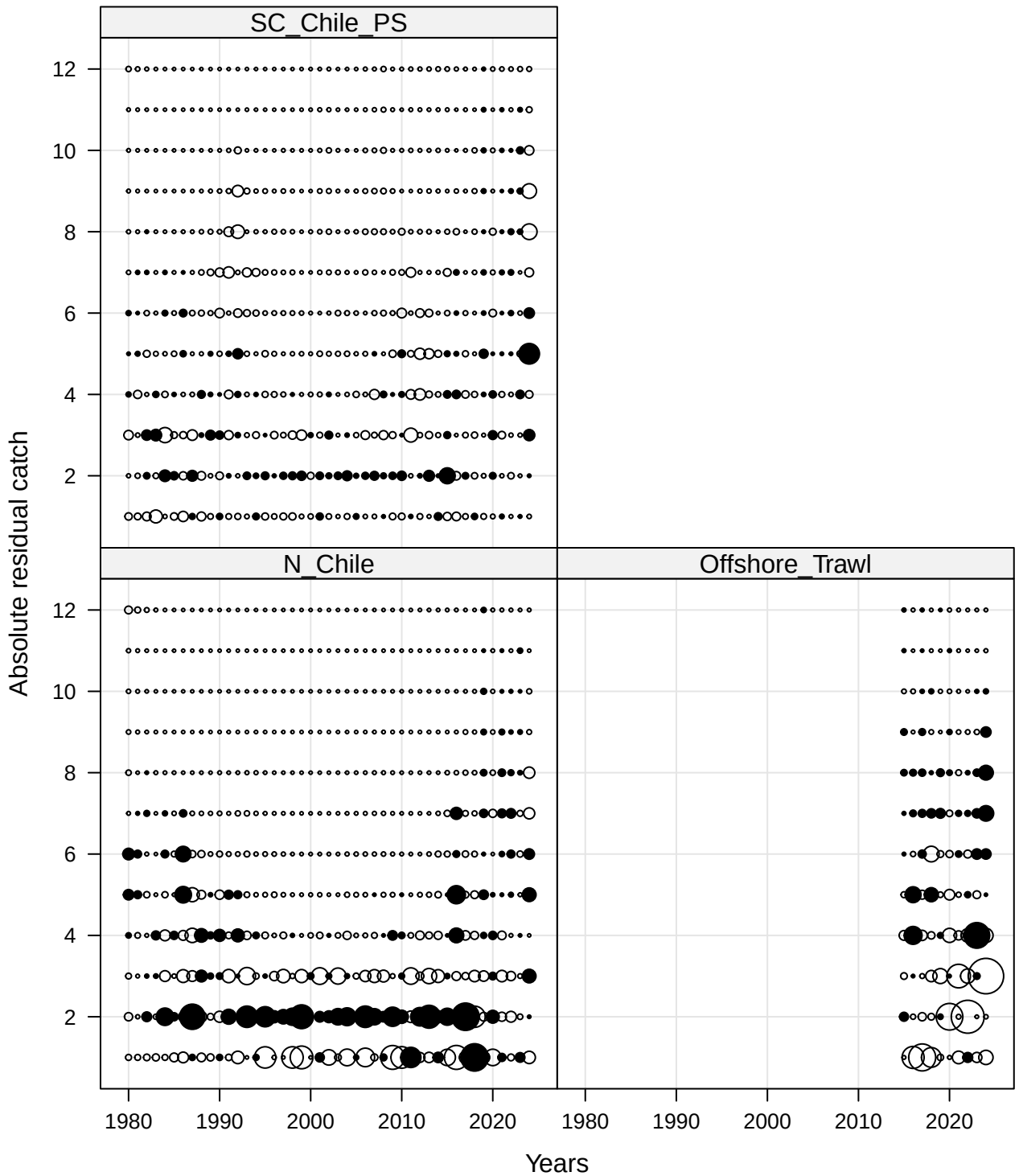
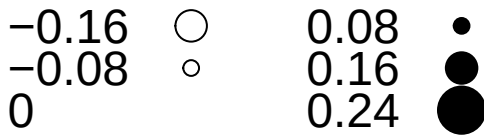


Catch residuals by fleet

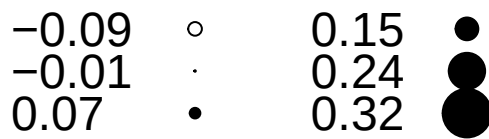




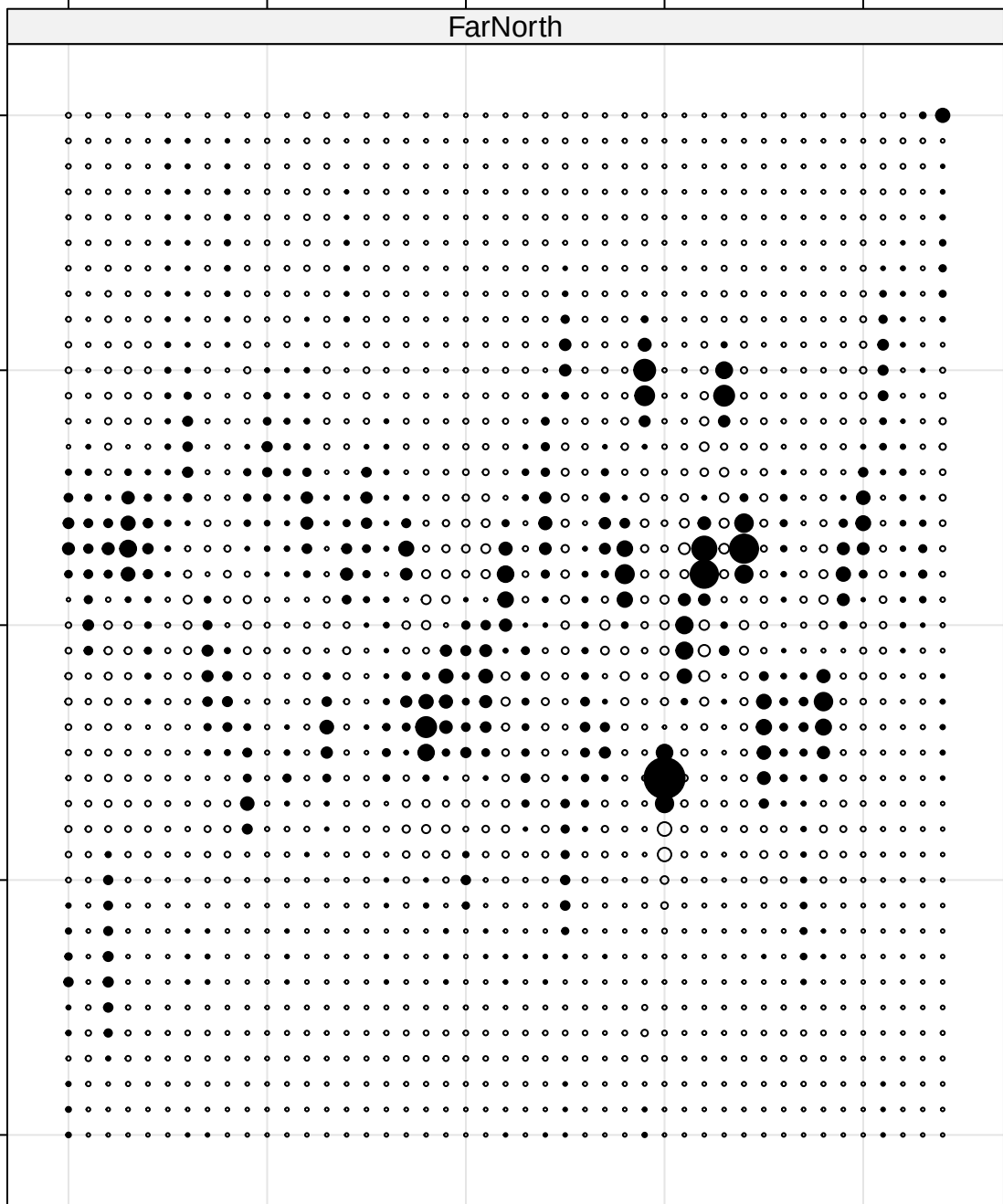
Absolute residual catch by fleet



Residuals catch-at-length by fleet



1980 1990 2000 2010 2020



1980 1990 2000 2010 2020

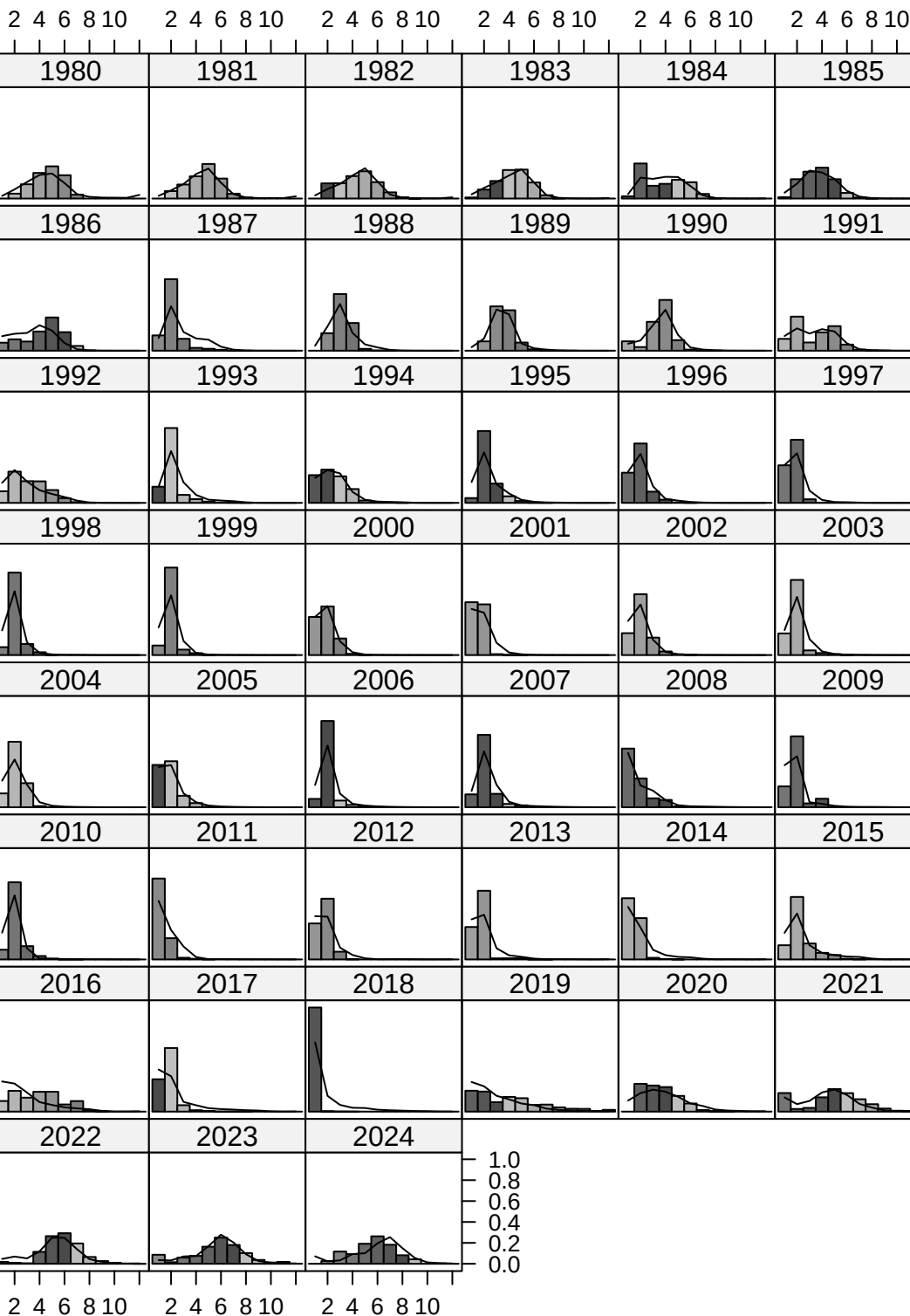
Years

Age fits N_Chile

Observed



Predicted



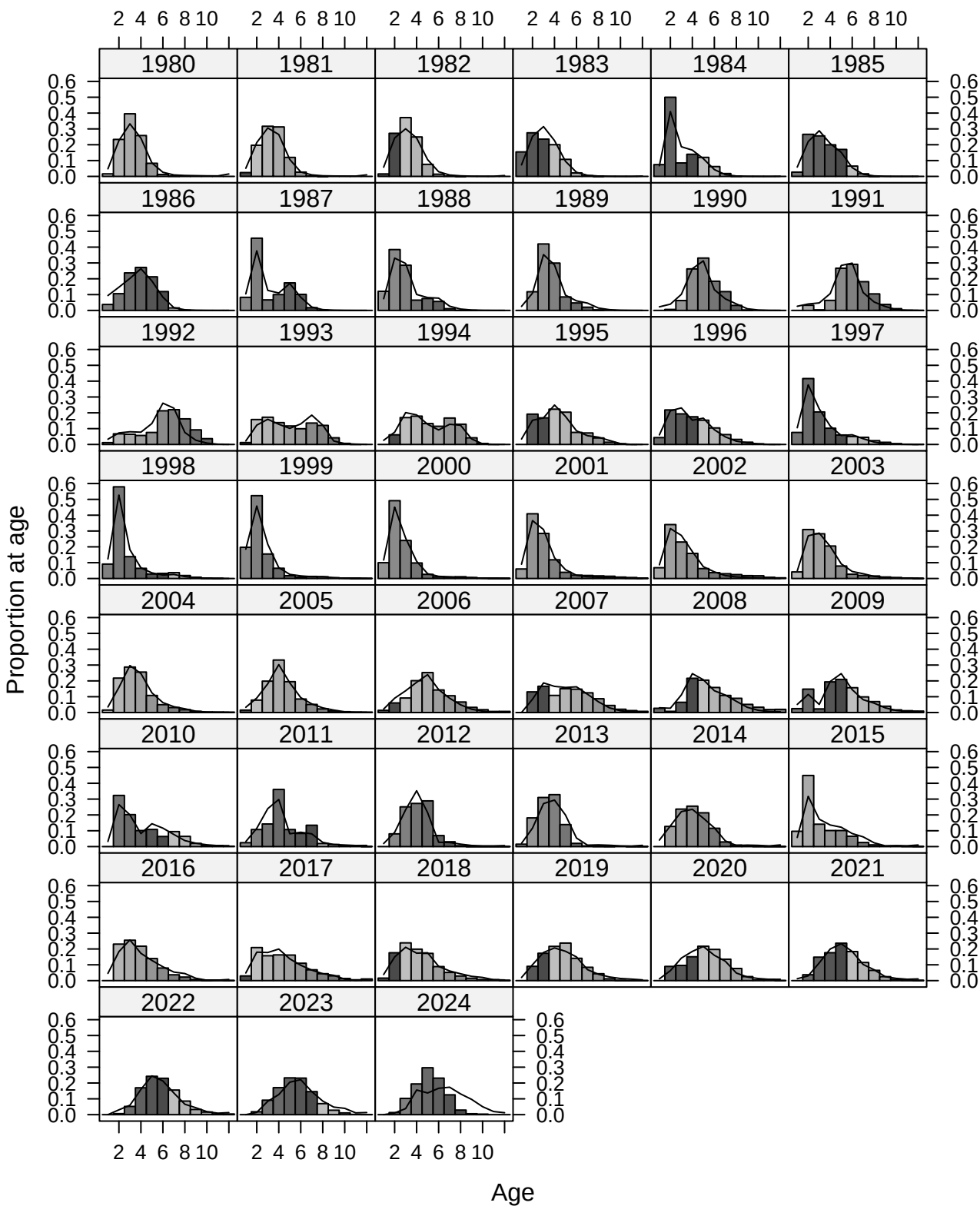
Age

Age fits SC_Chile_PS

Observed



Predicted ●



Age fits Offshore_Trawl

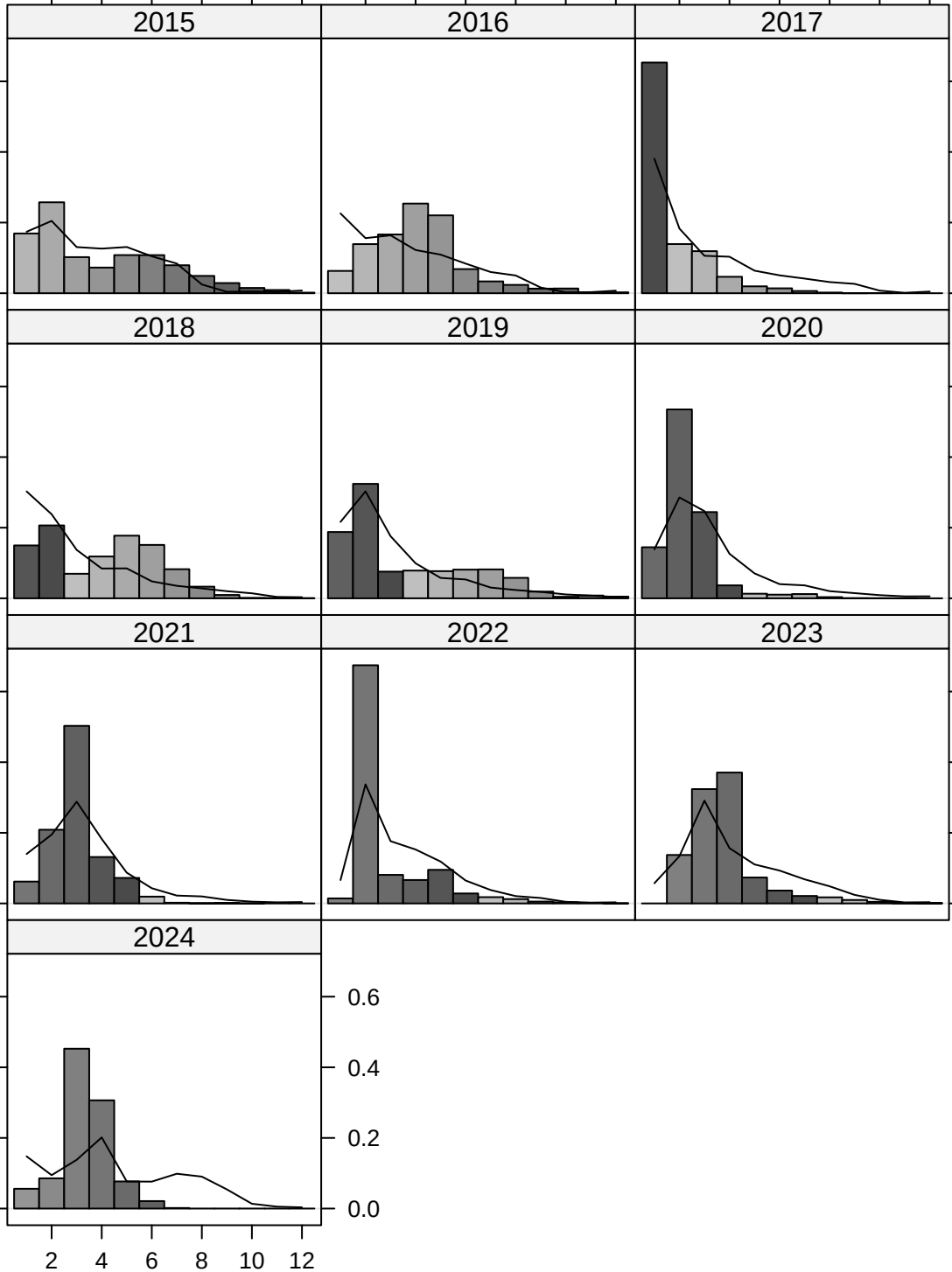
Observed



Predicted



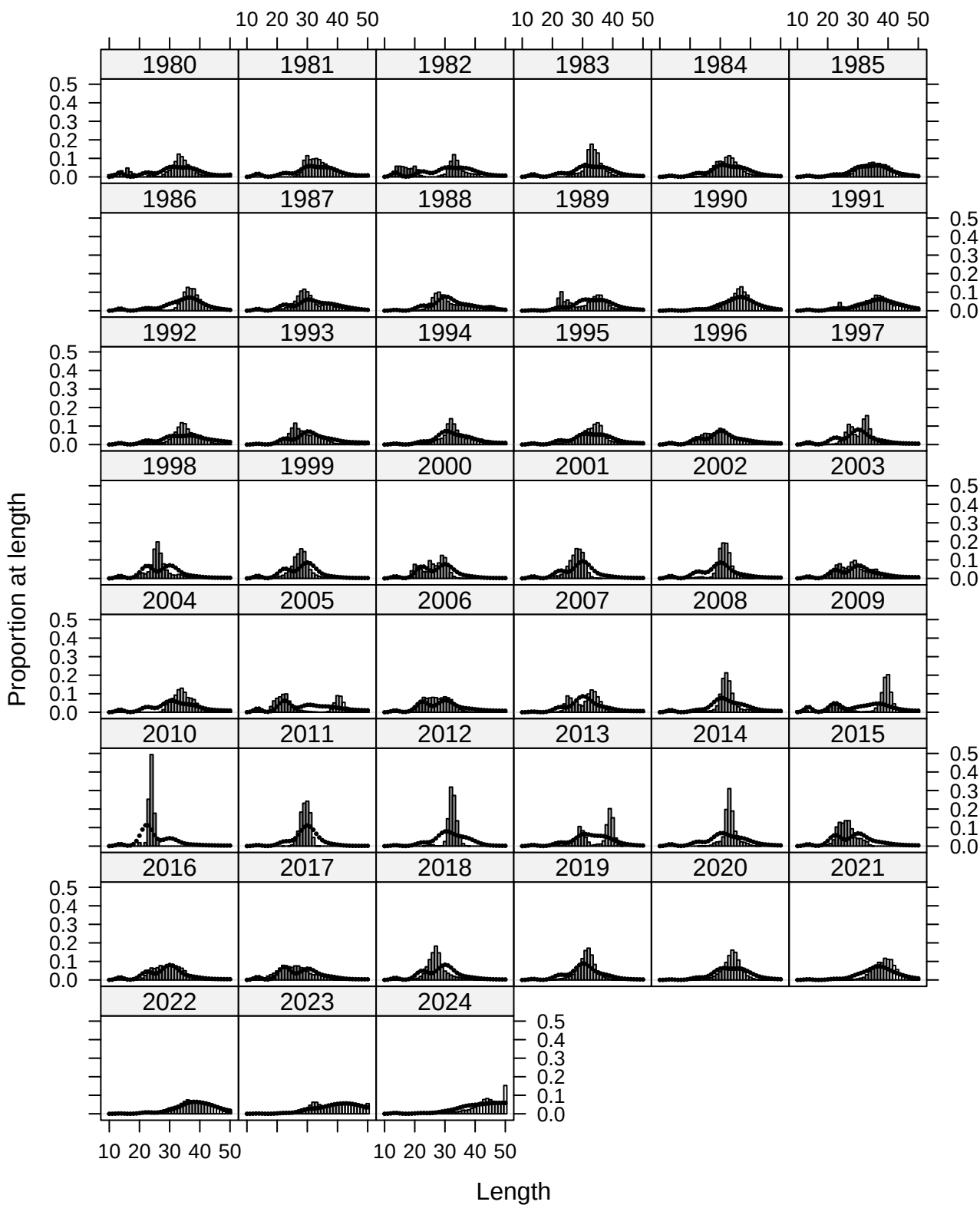
2 4 6 8 10 12 2 4 6 8 10 12 2 4 6 8 10 12



Age

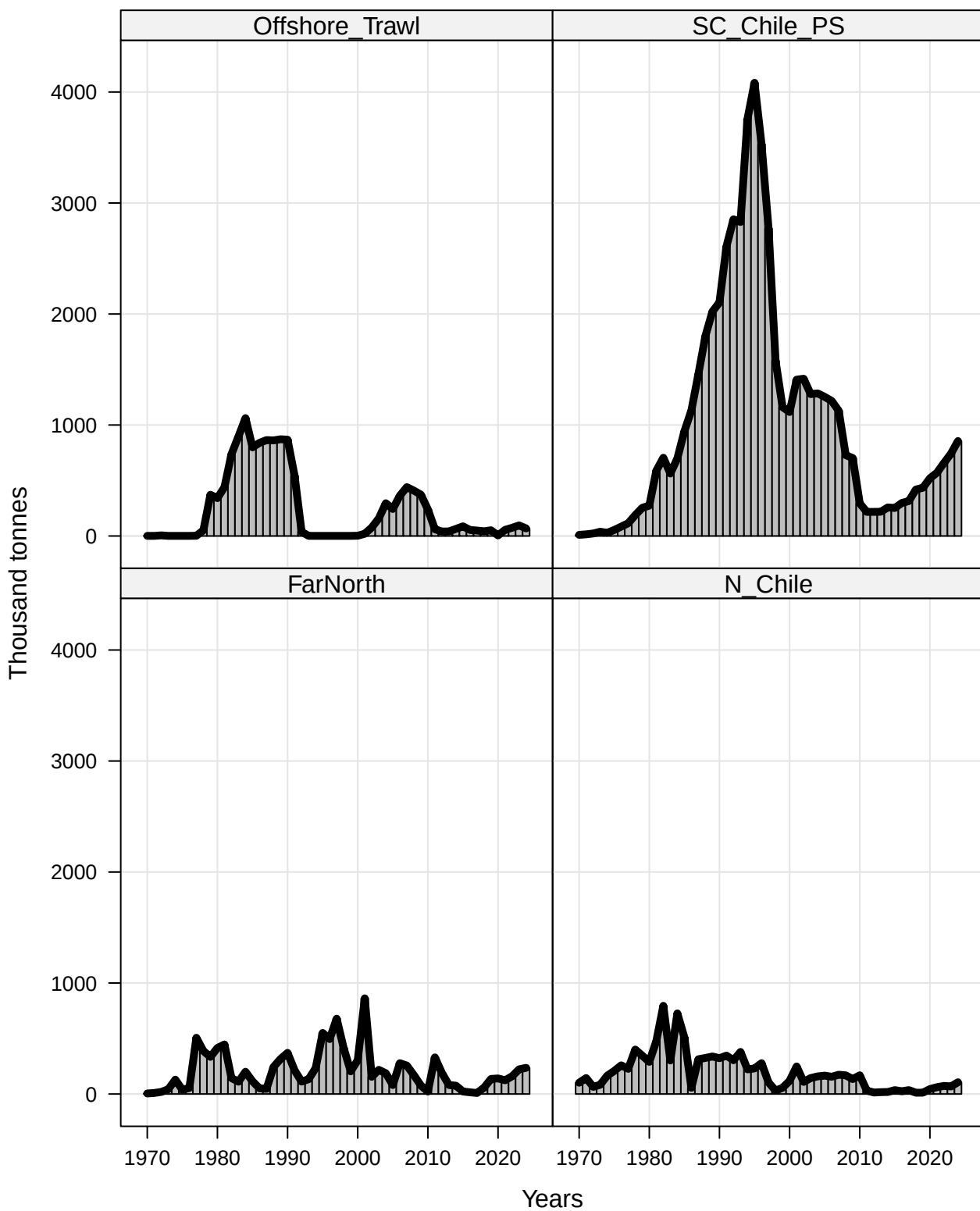
Length fits FarNorth

Observed  Predicted 



Predicted and observed catches by fleet

Observed Predicted

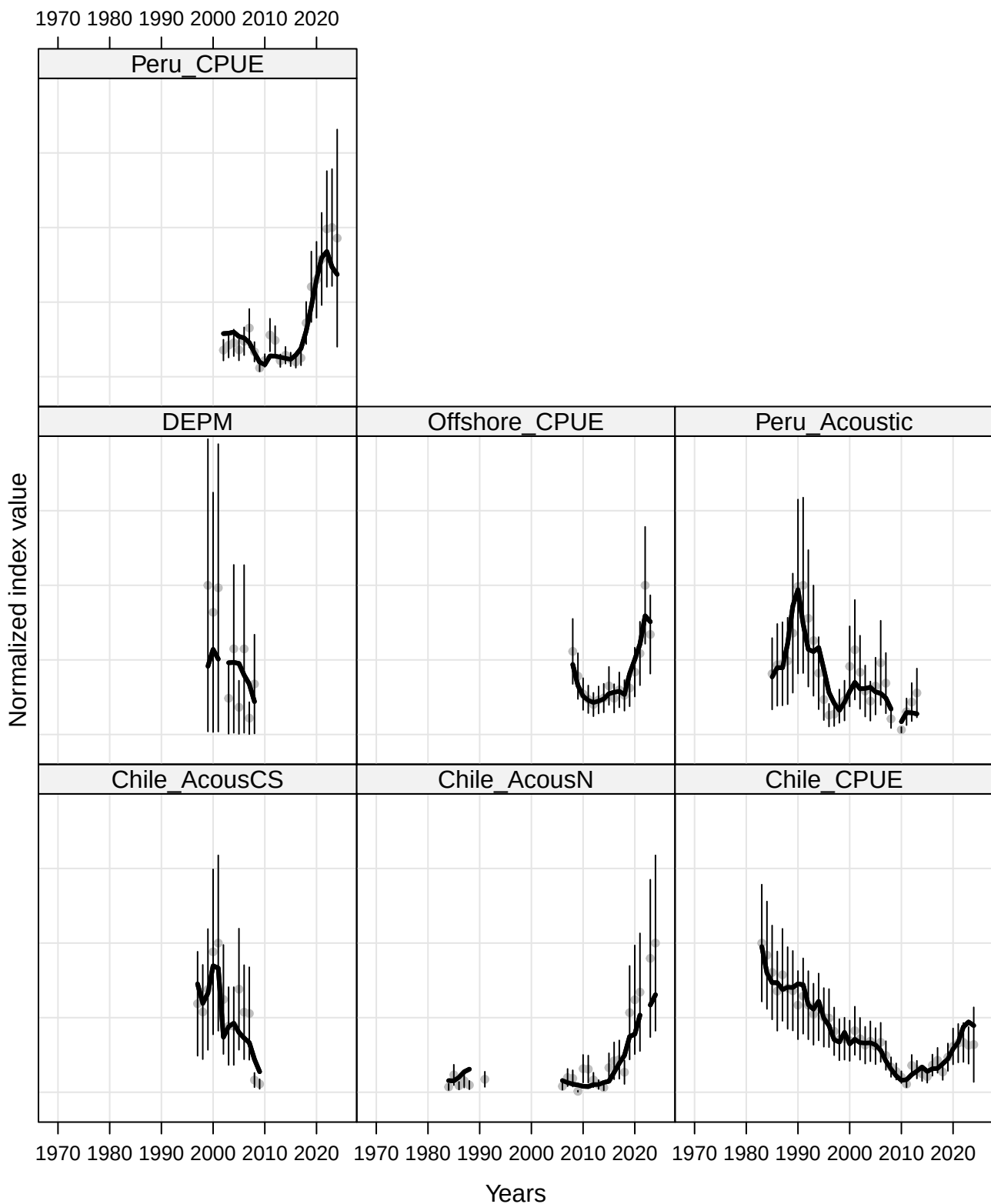


Predicted and observed indices

Observed



Predicted



Age fits Chile_AcoustCS

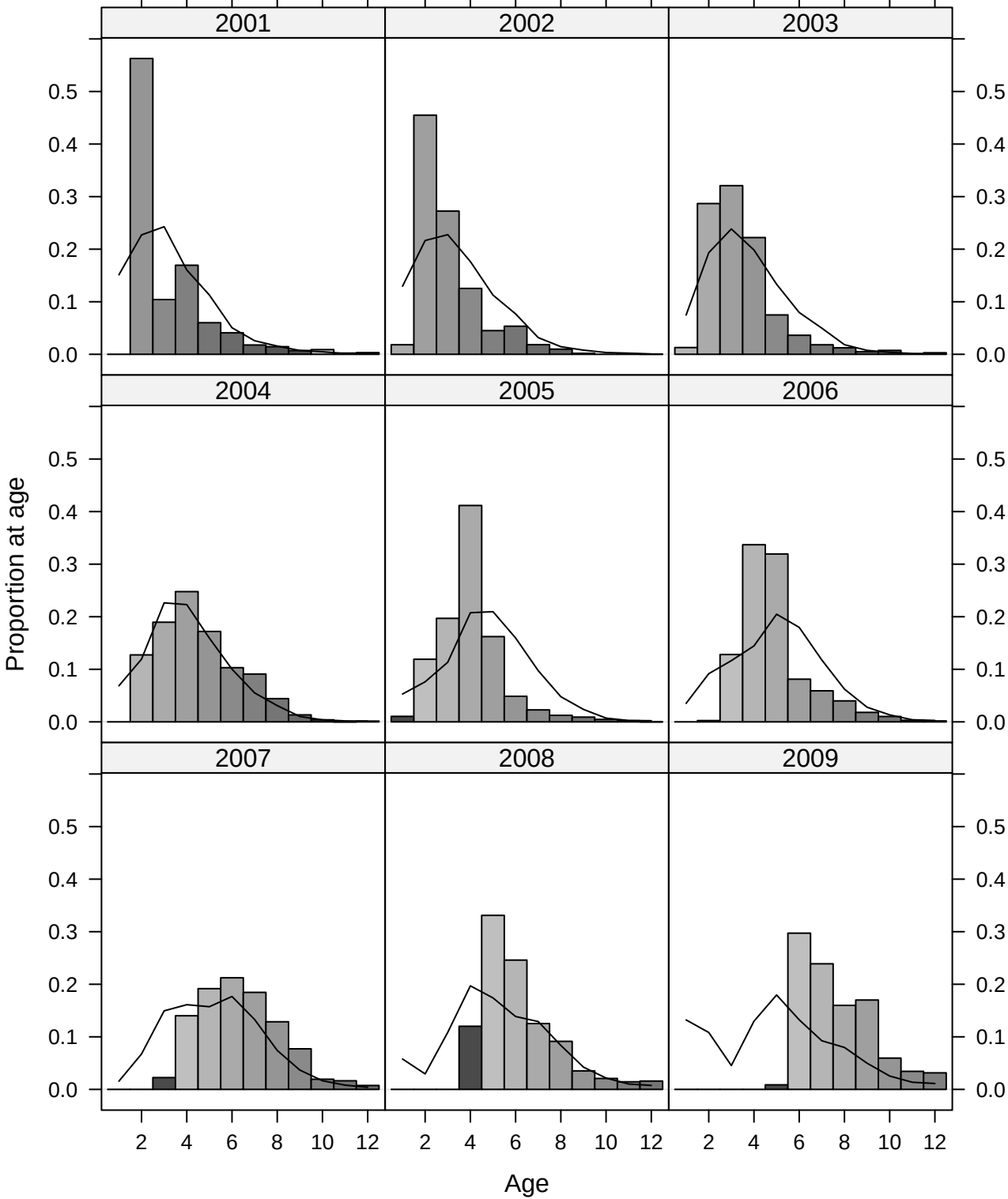
Observed



Predicted



2 4 6 8 10 12 2 4 6 8 10 12 2 4 6 8 10 12



Age fits Chile_AcousN

Observed



Predicted



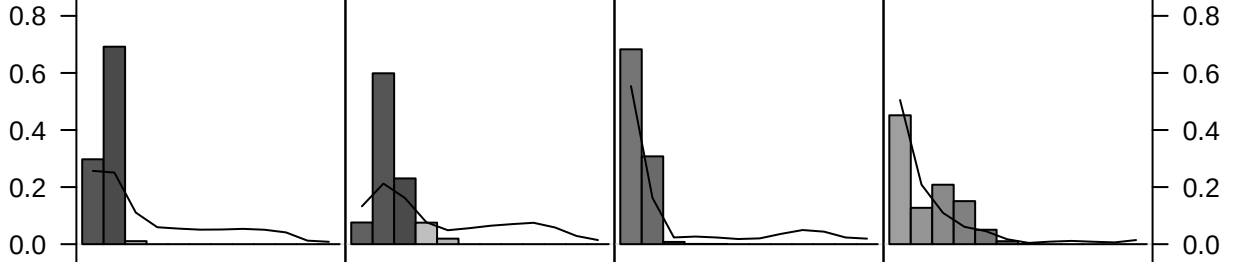
2 4 6 8 10 12 2 4 6 8 10 12 2 4 6 8 10 12 2 4 6 8 10 12

2006

2007

2009

2013

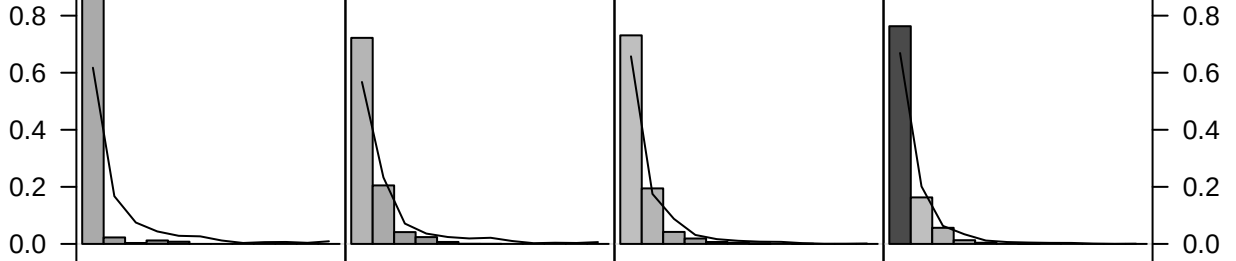


2014

2015

2016

2017

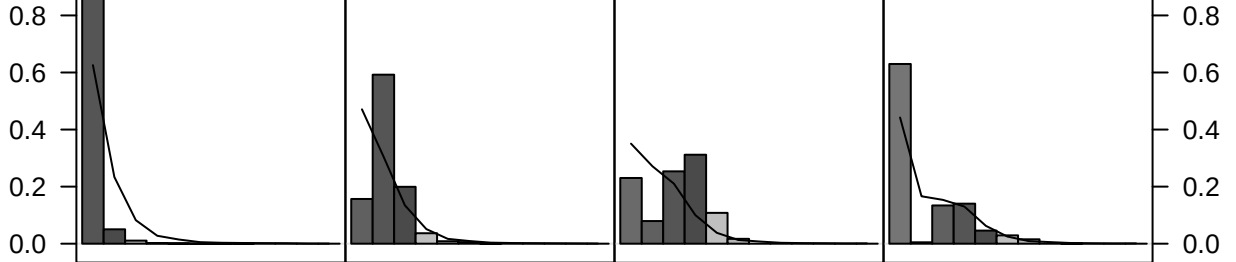


2018

2019

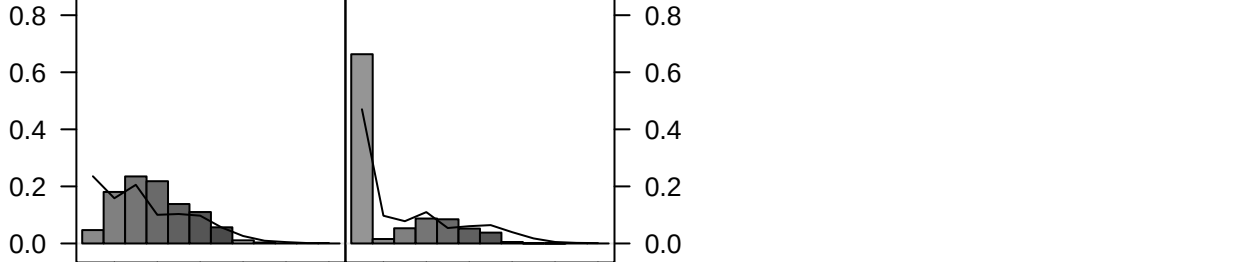
2020

2021



2023

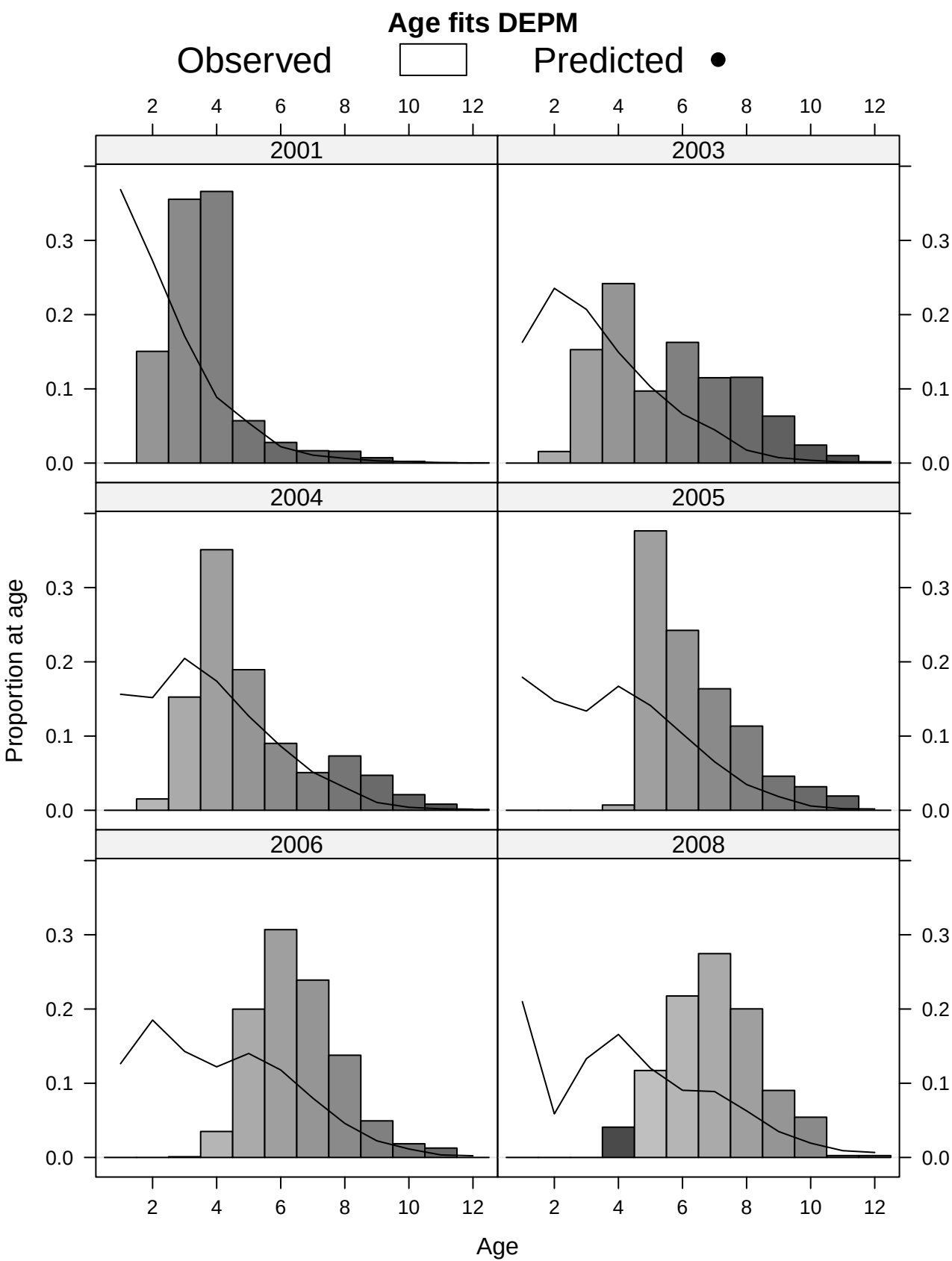
2024



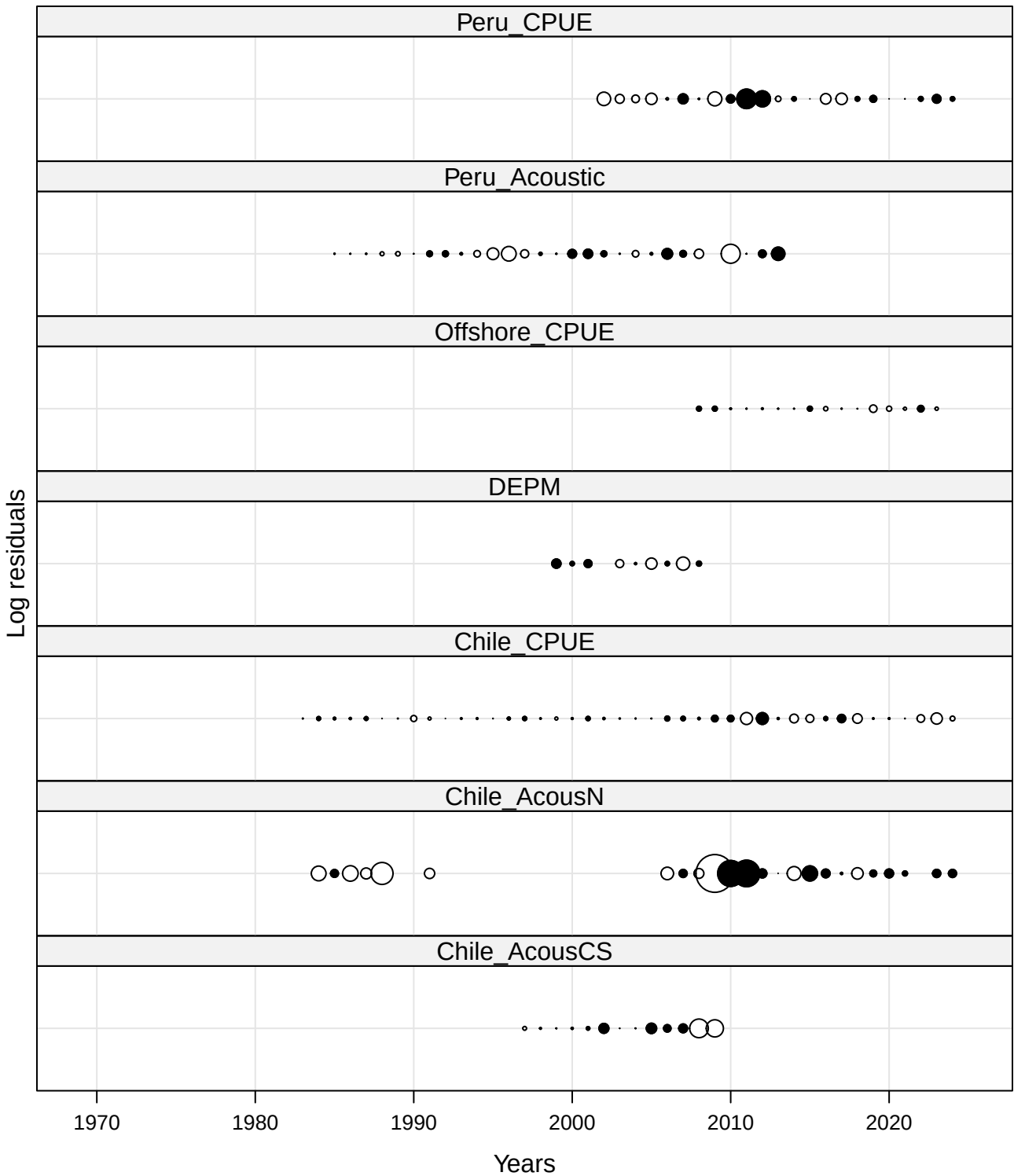
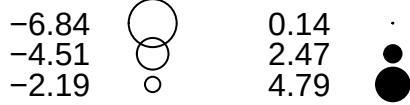
2 4 6 8 10 12 2 4 6 8 10 12

Age

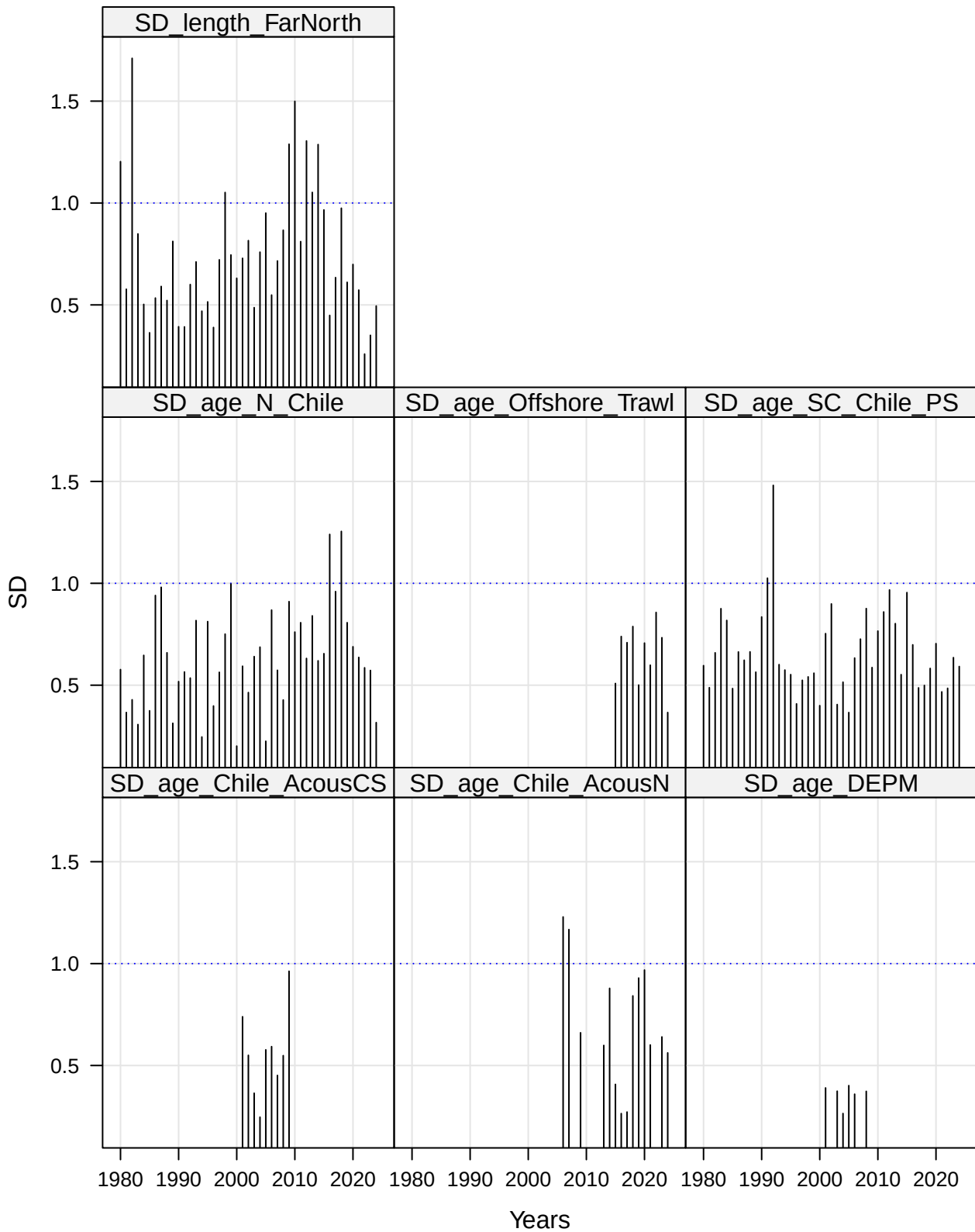
Proportion at age



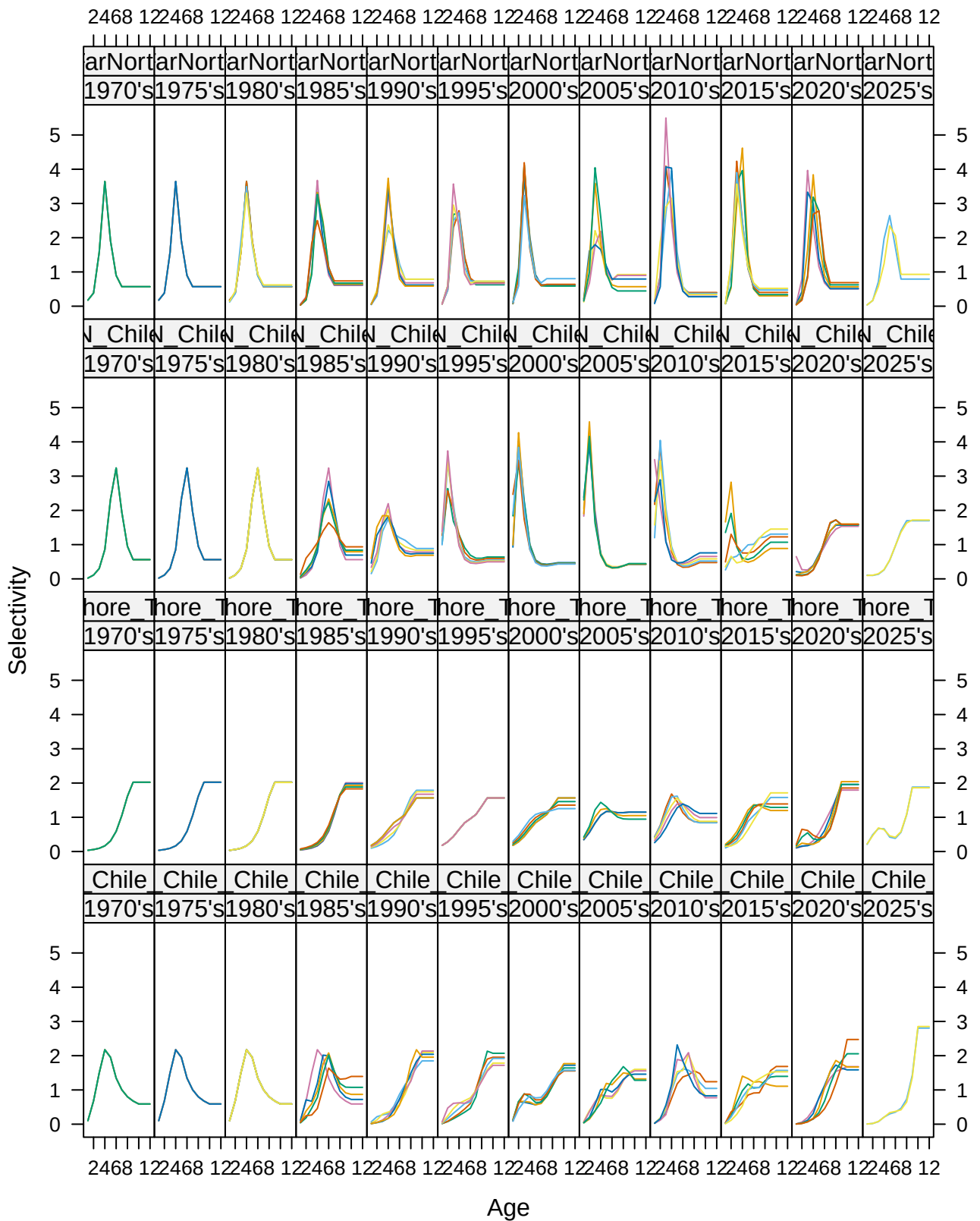
Standardized survey residuals



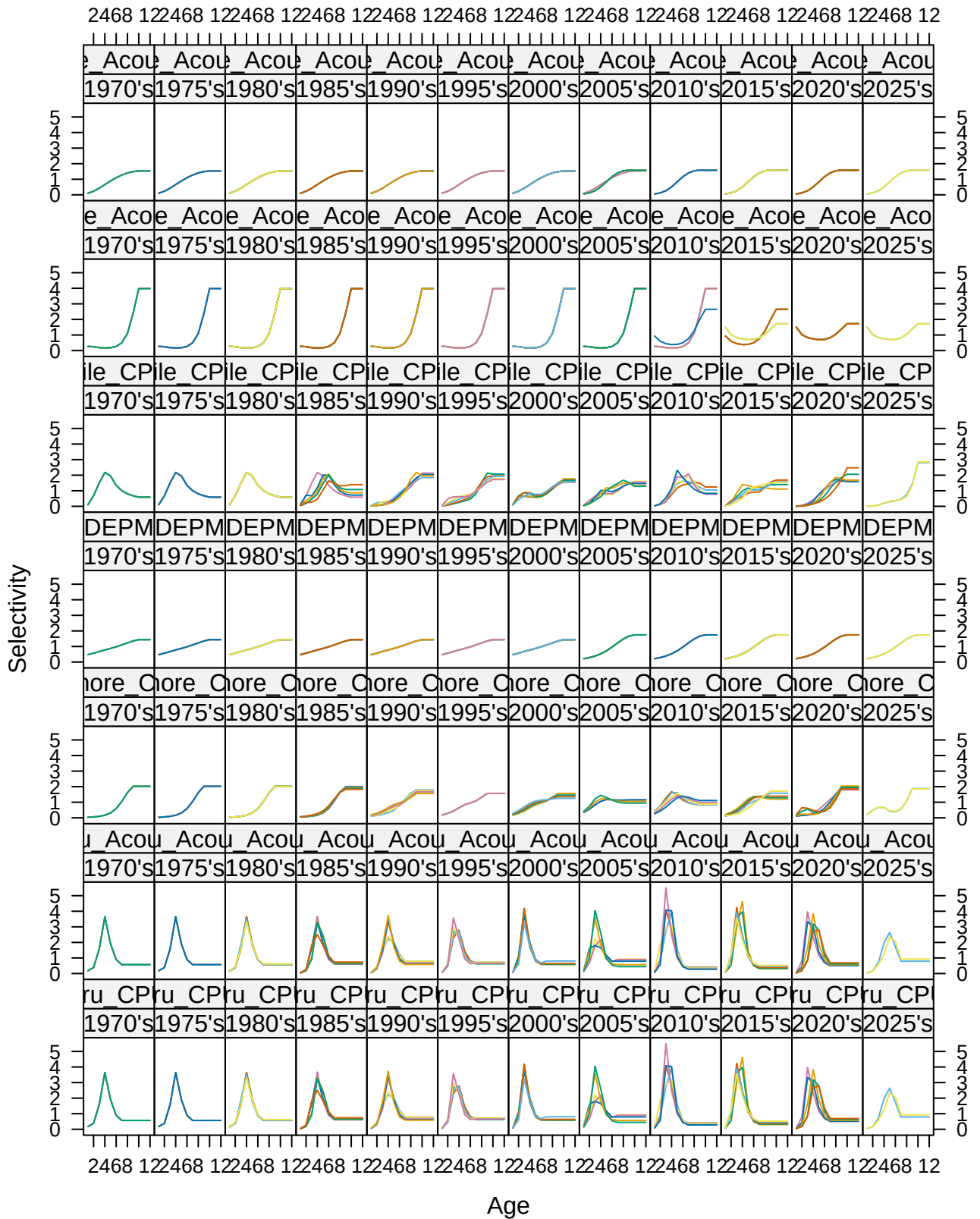
SD per input series



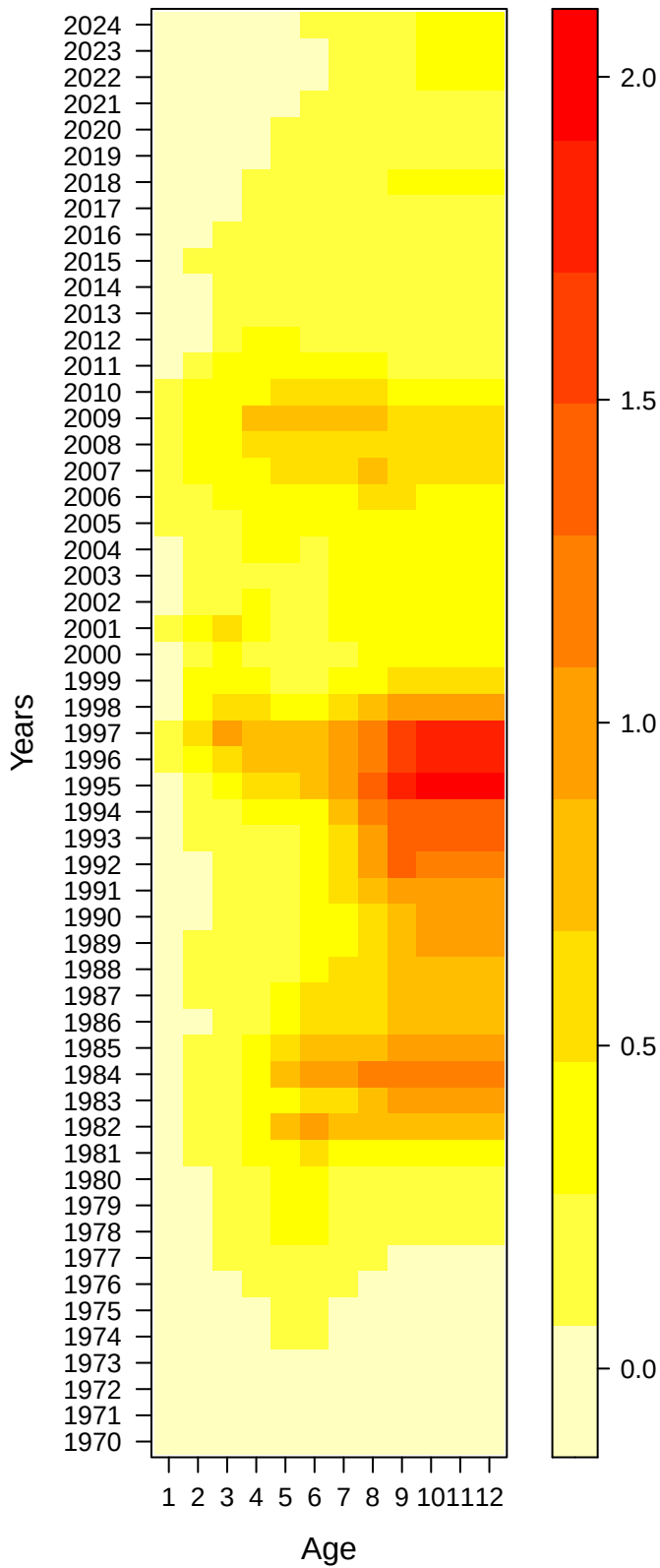
Selectivity of the Fishery by Pentad



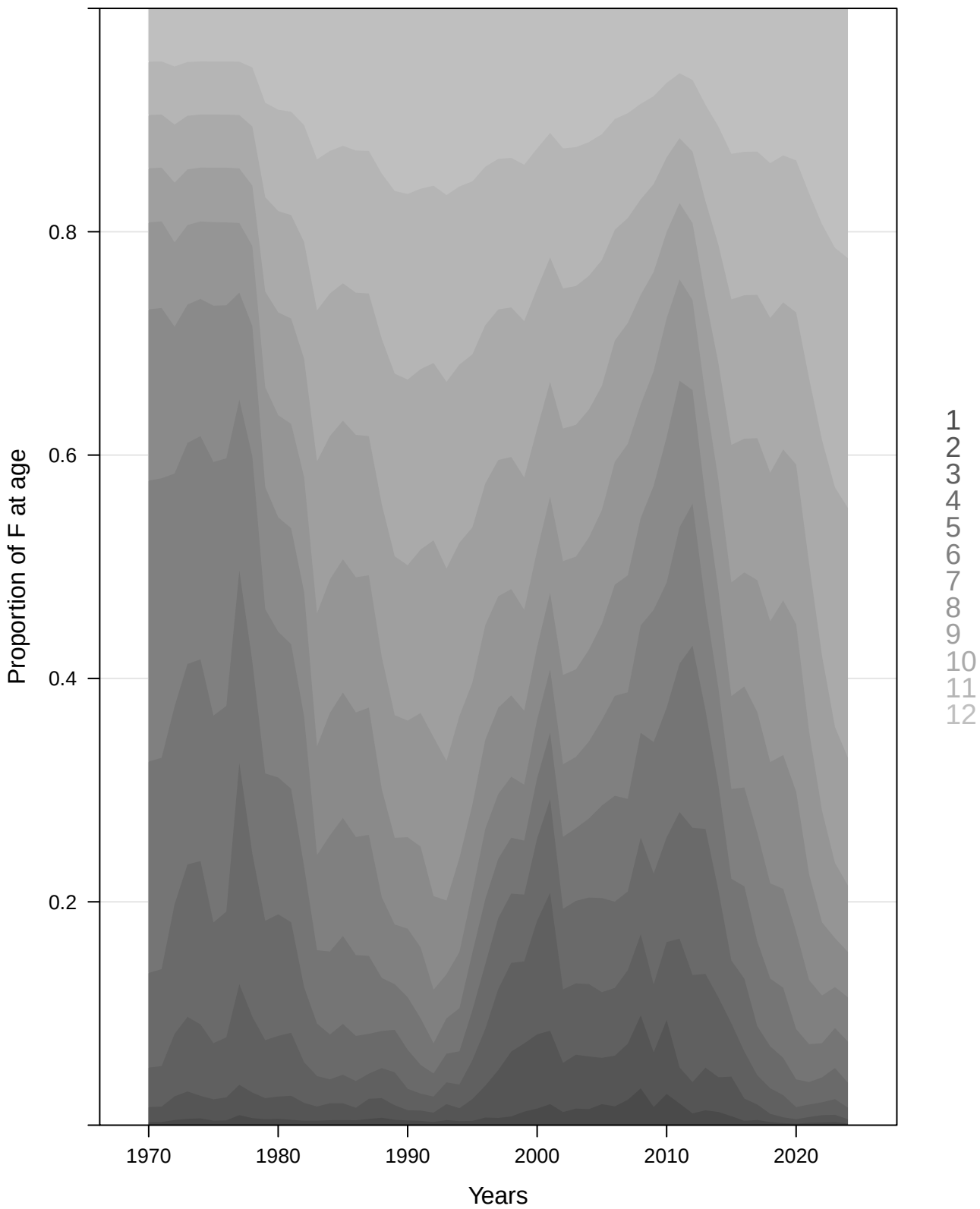
Selectivity of the survey by Pentad



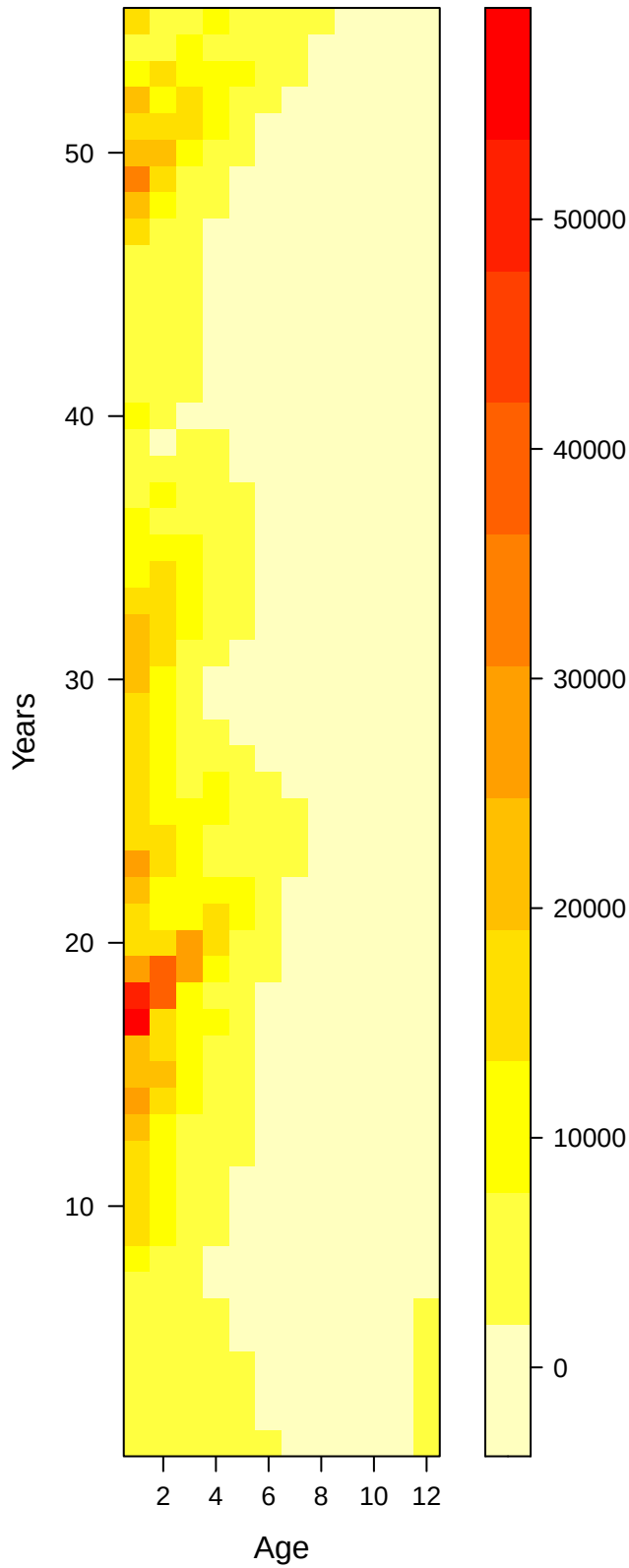
F at age



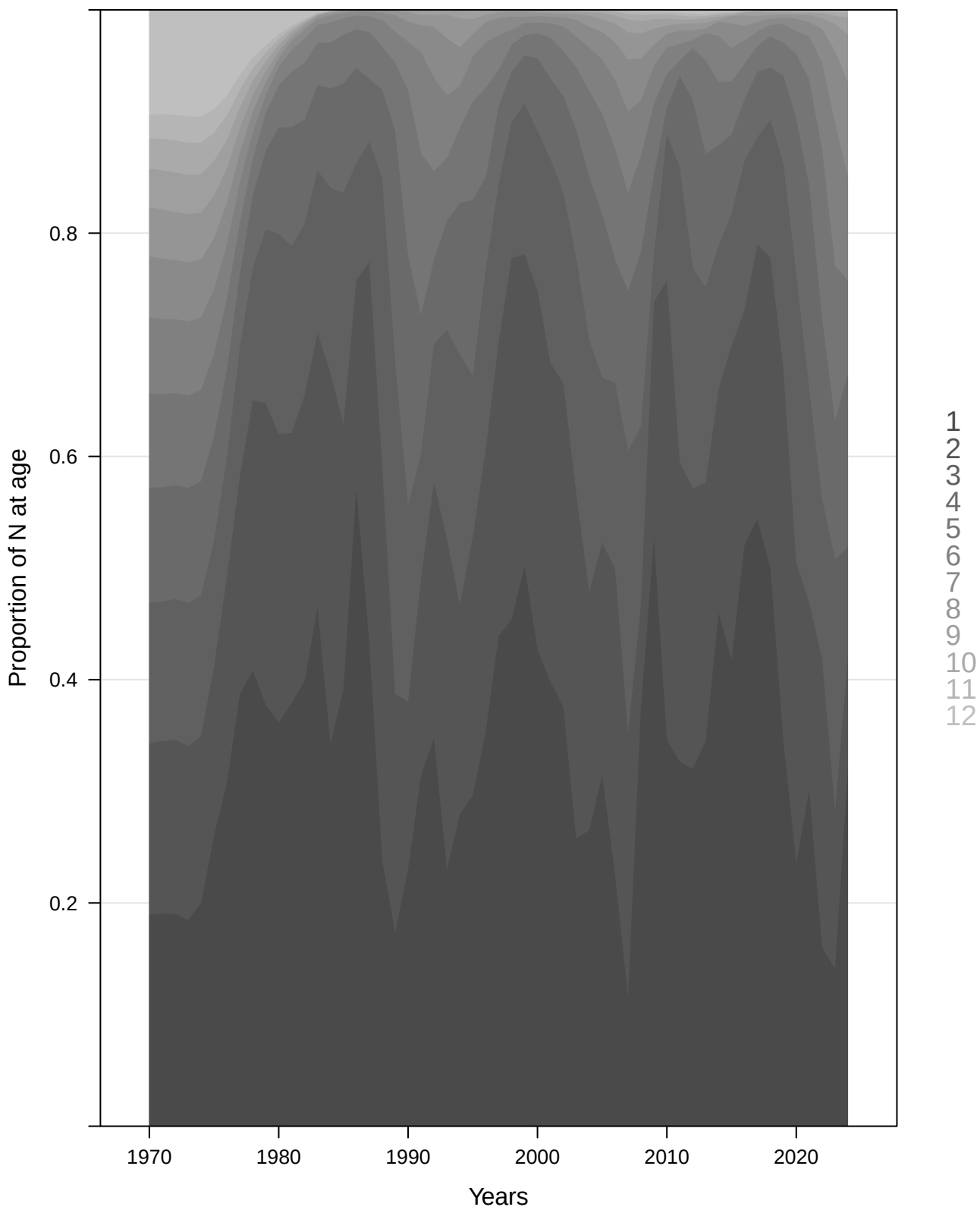
F proportion at age



N at age



Proportion at age

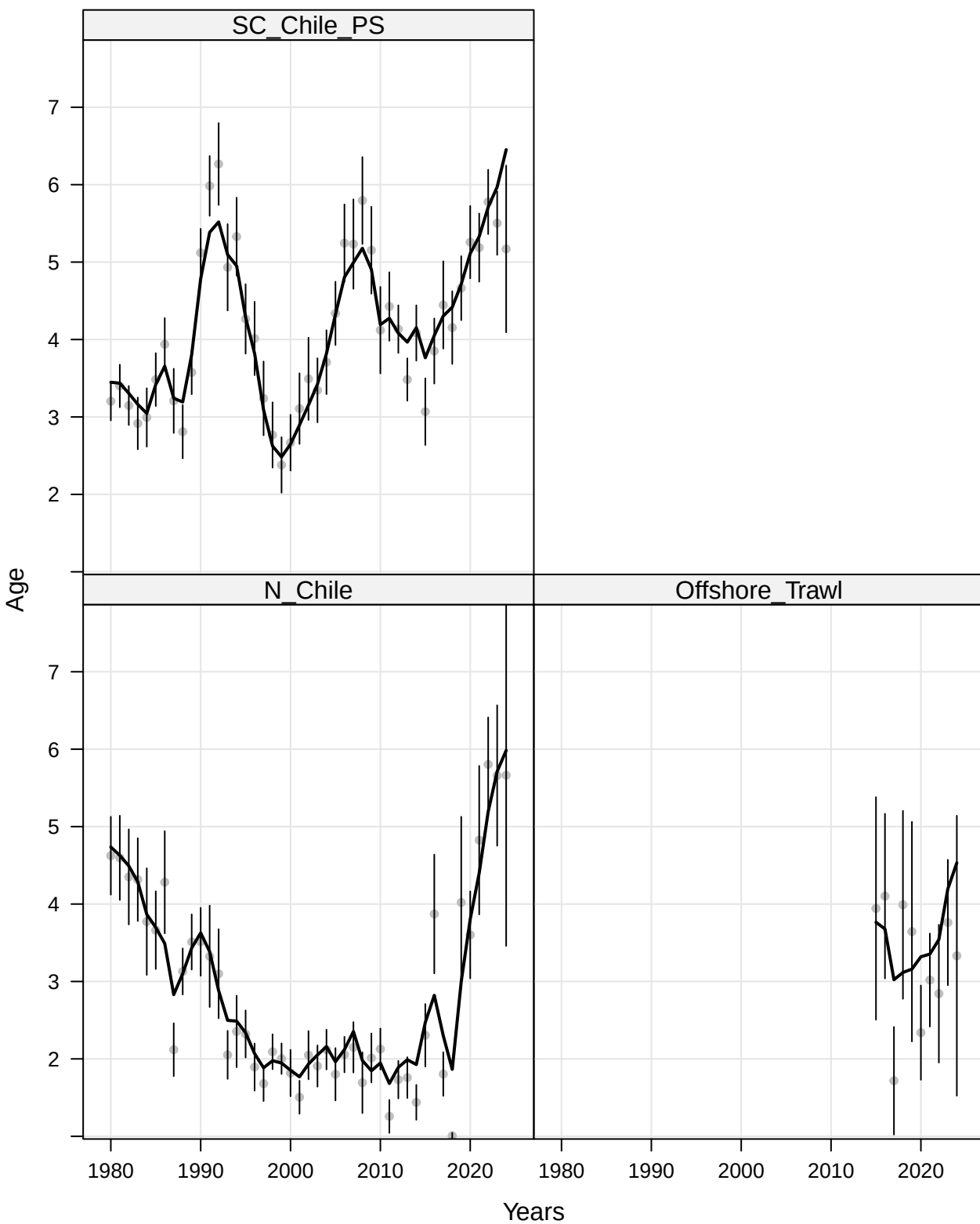


Fishery mean age

Observed



Modelled

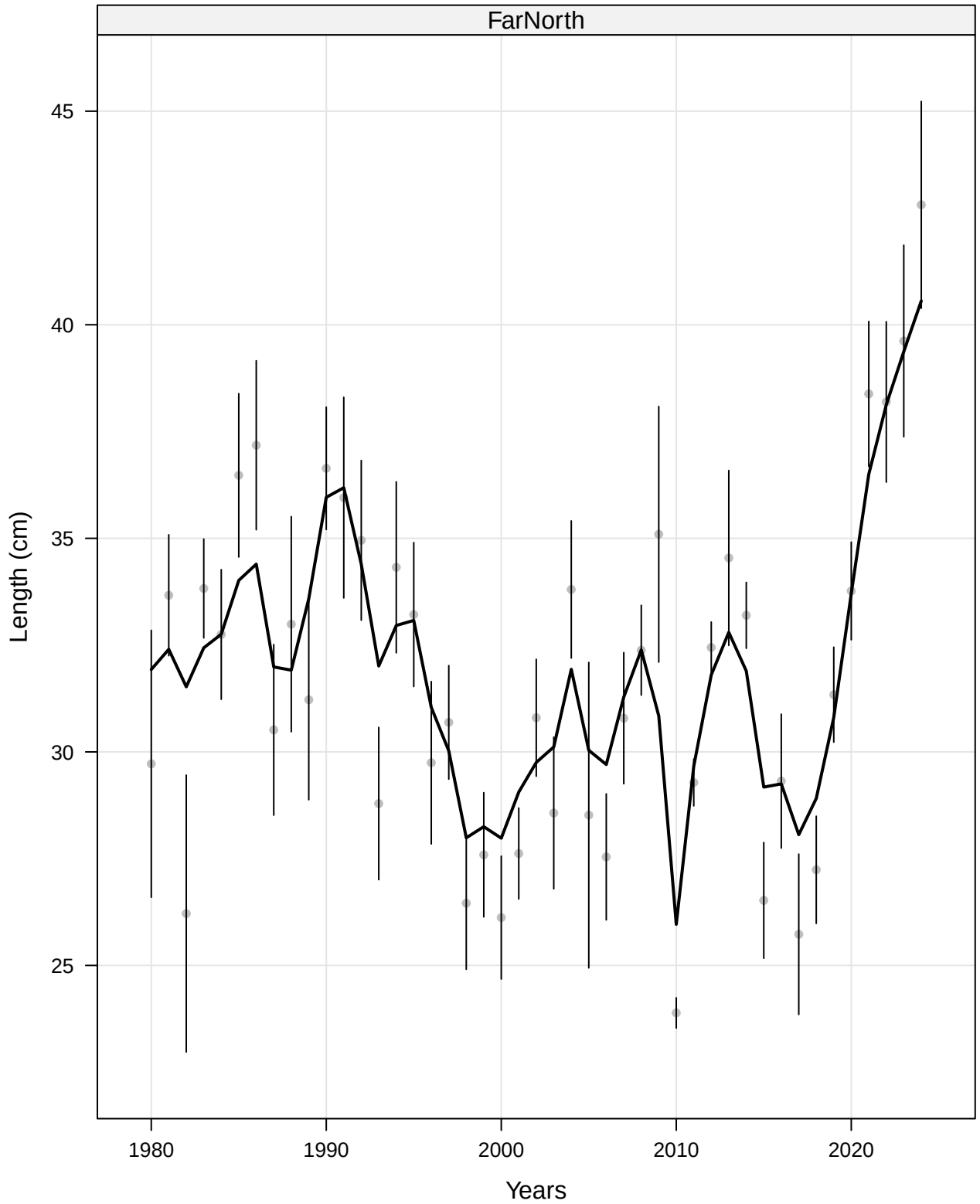


Fishery mean length

Observed



Modelled

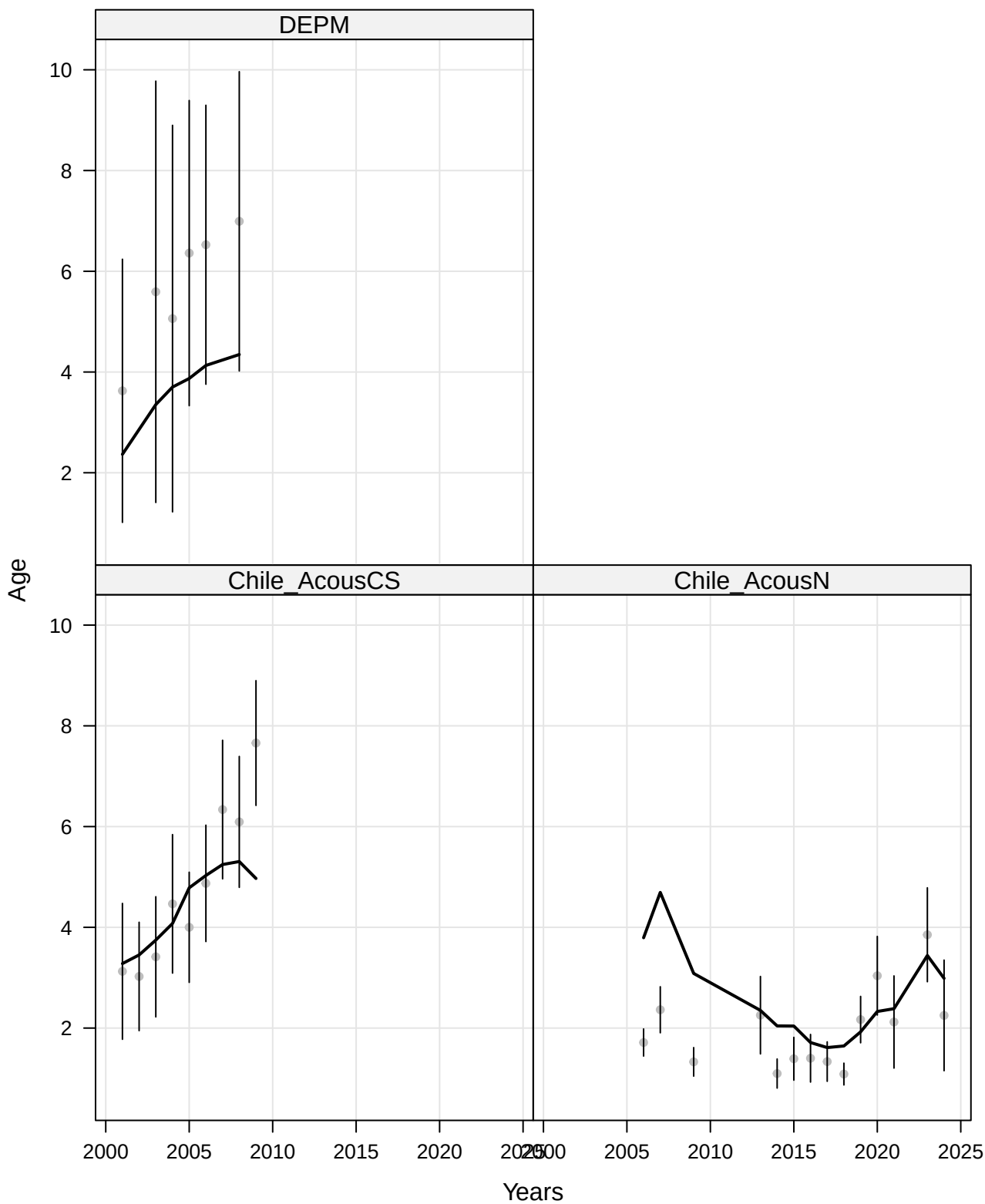


Survey mean age

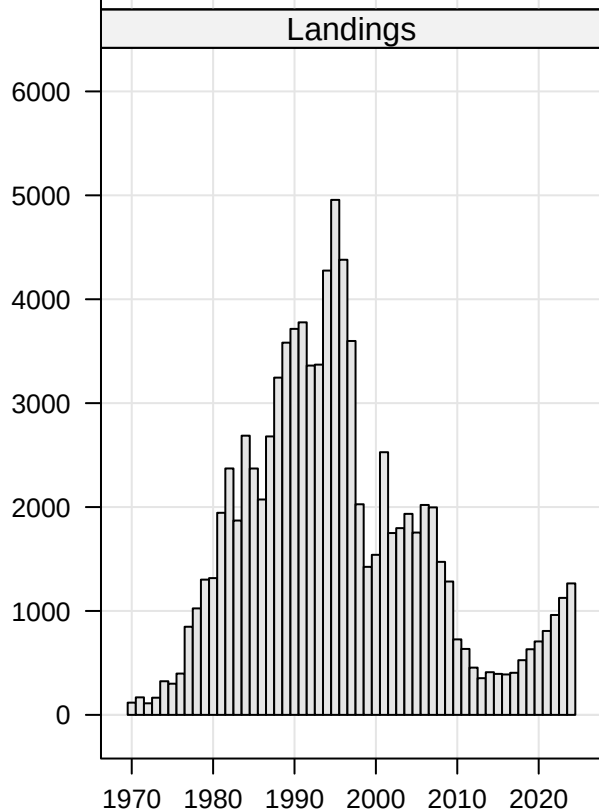
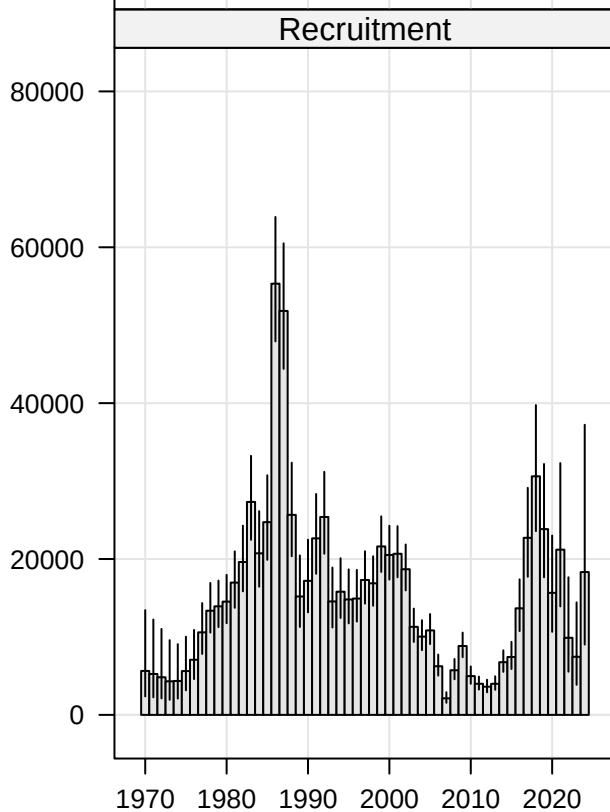
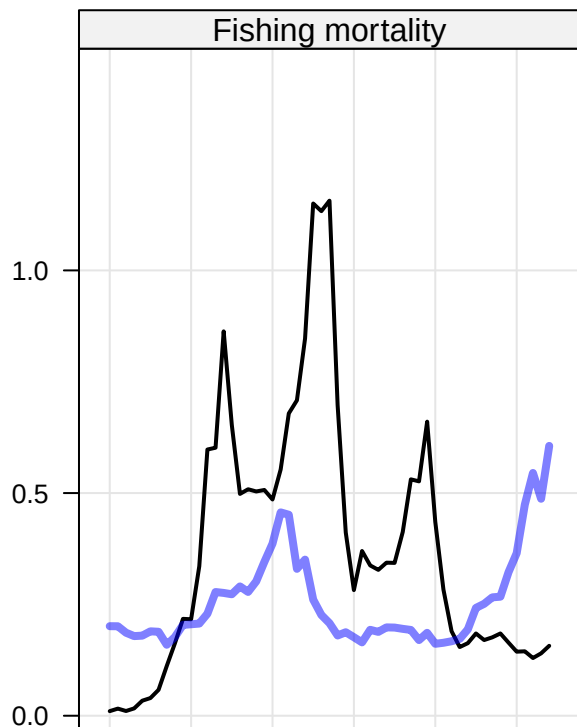
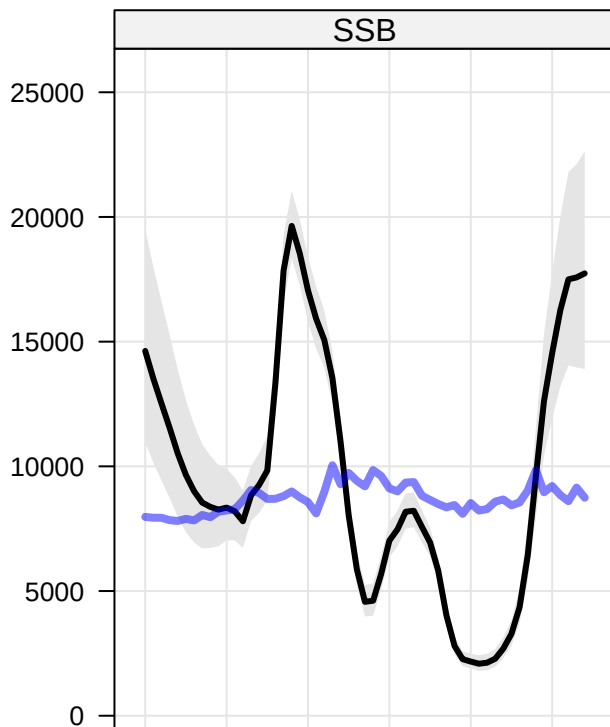
Observed



Modelled

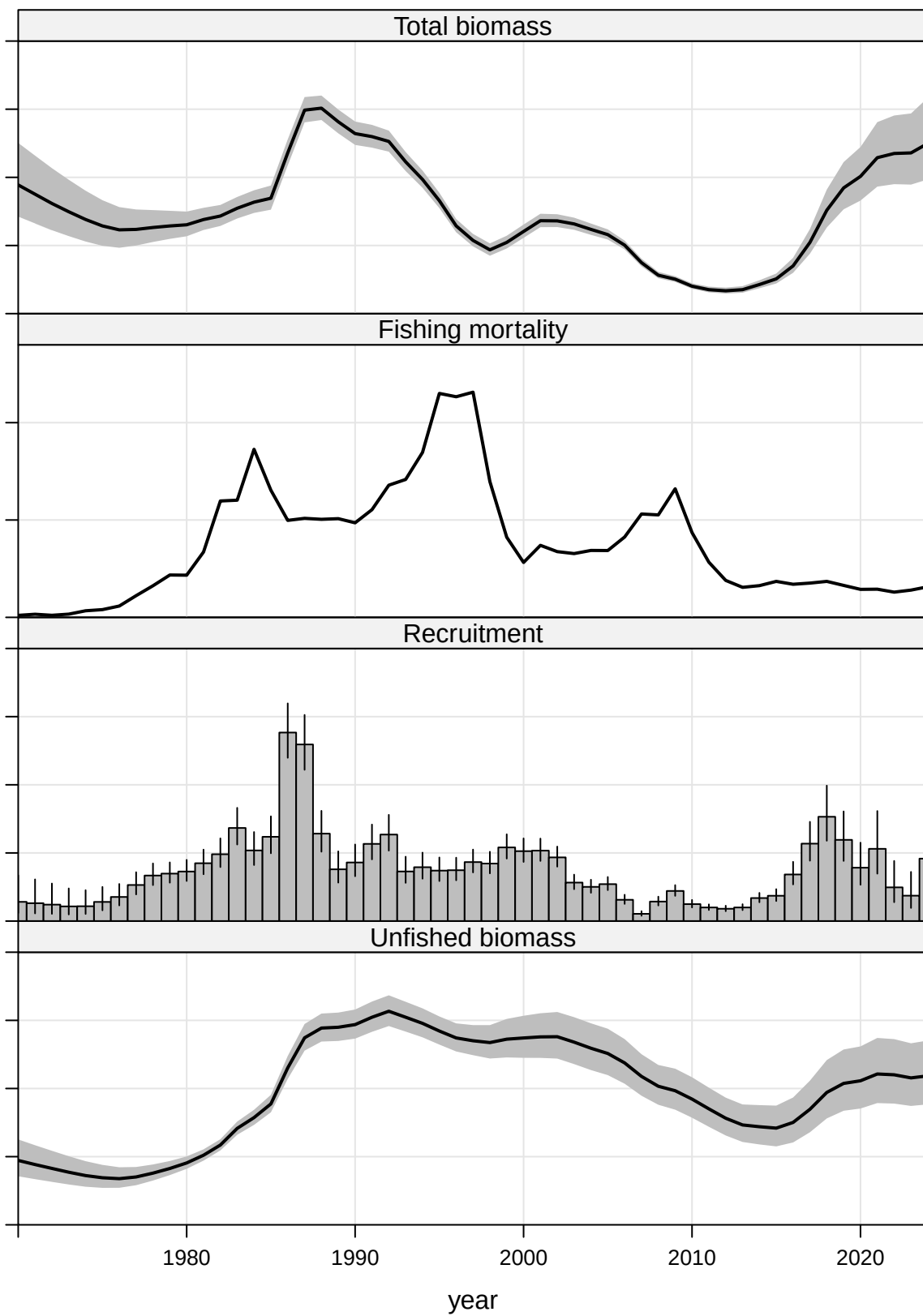


data

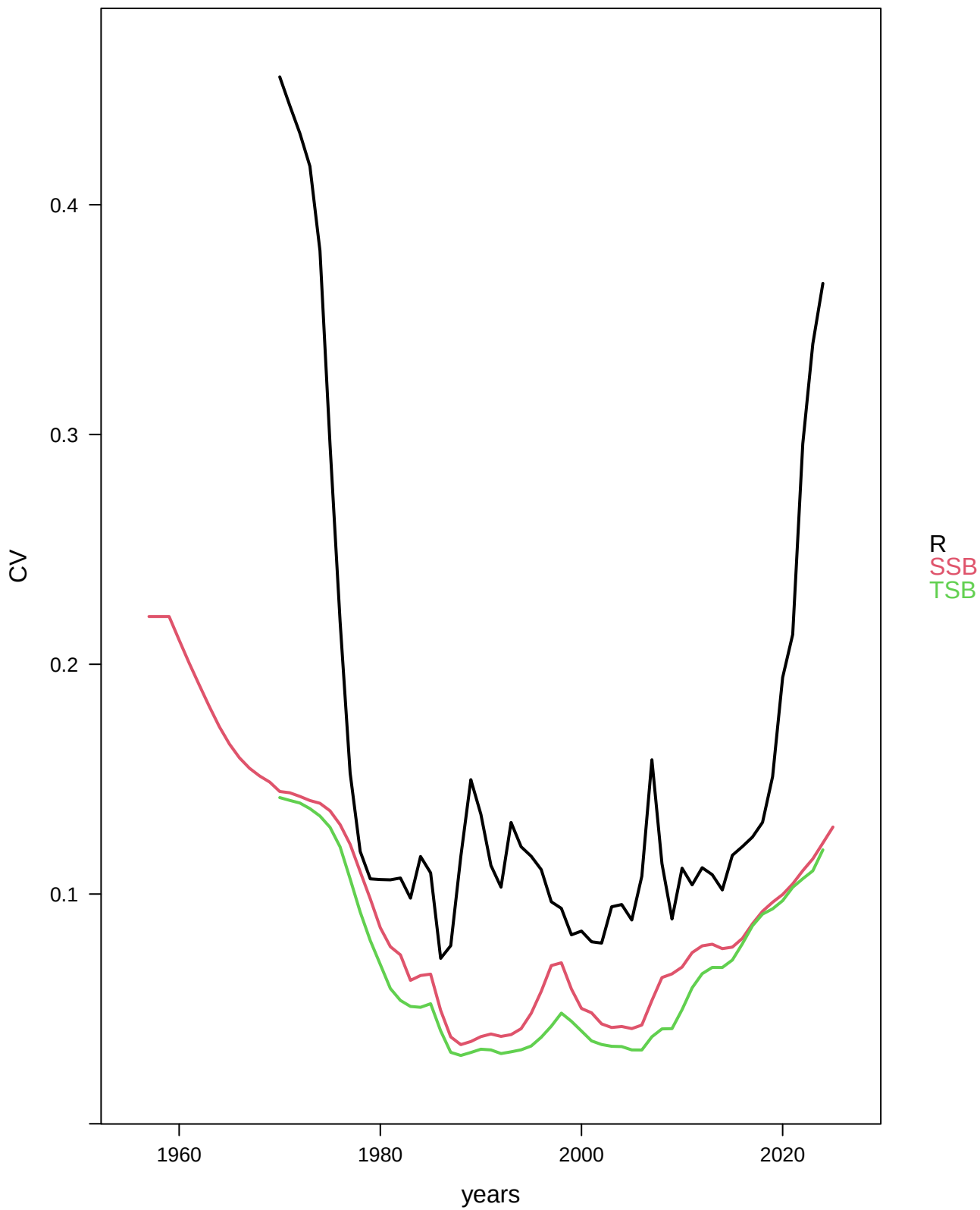


year

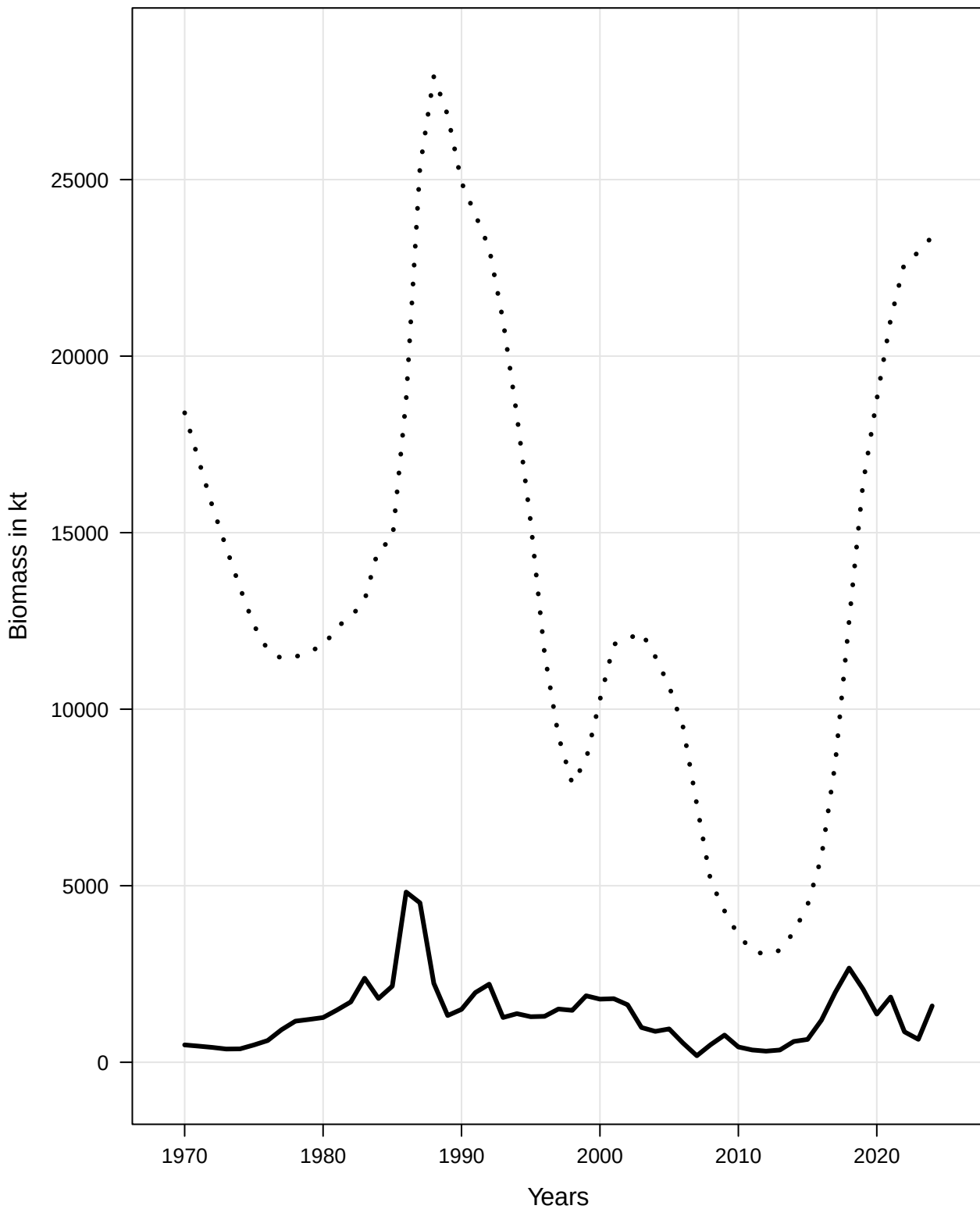
data



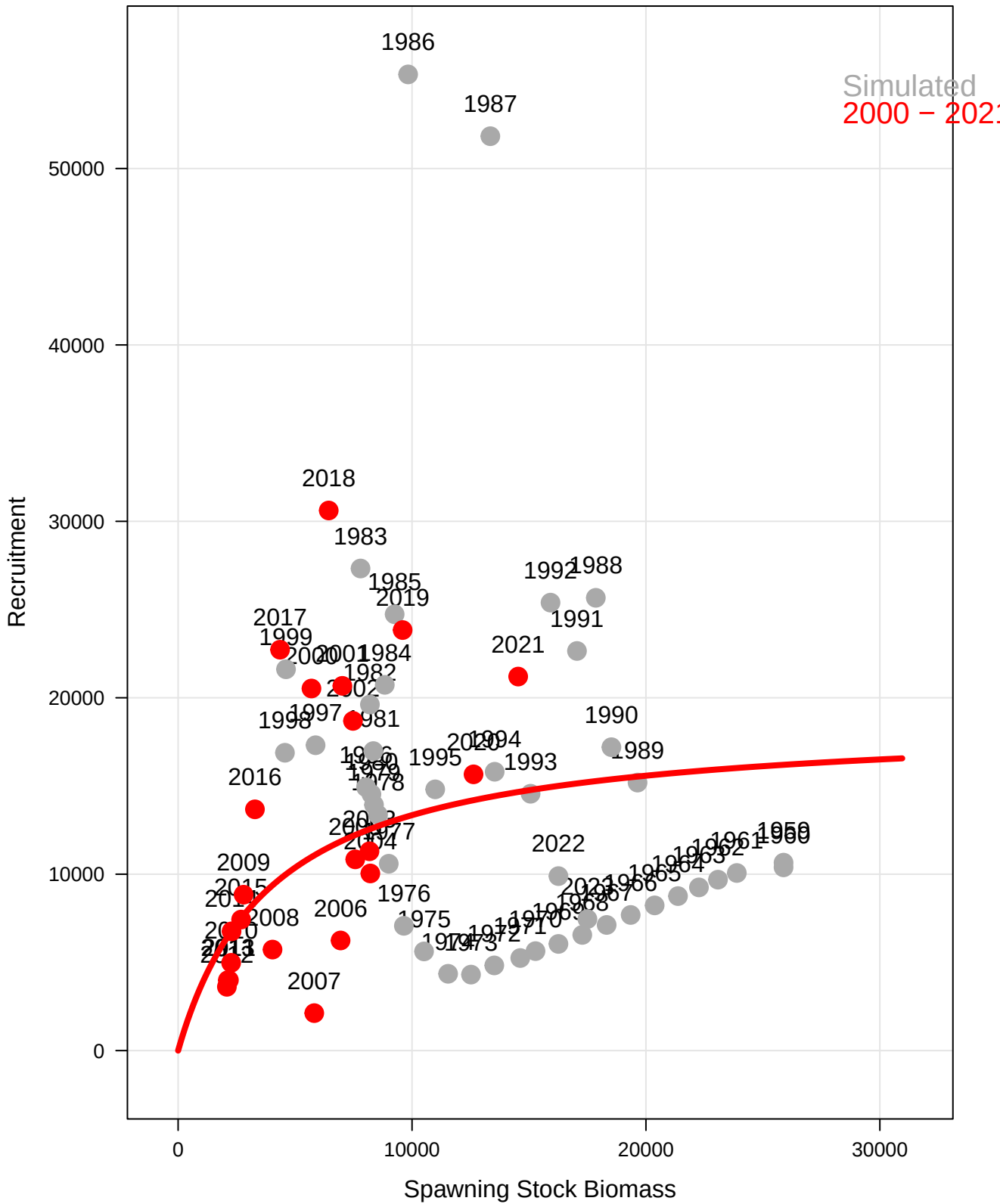
Uncertainty of key parameters



Mature – Immature fish
Mature Immature —

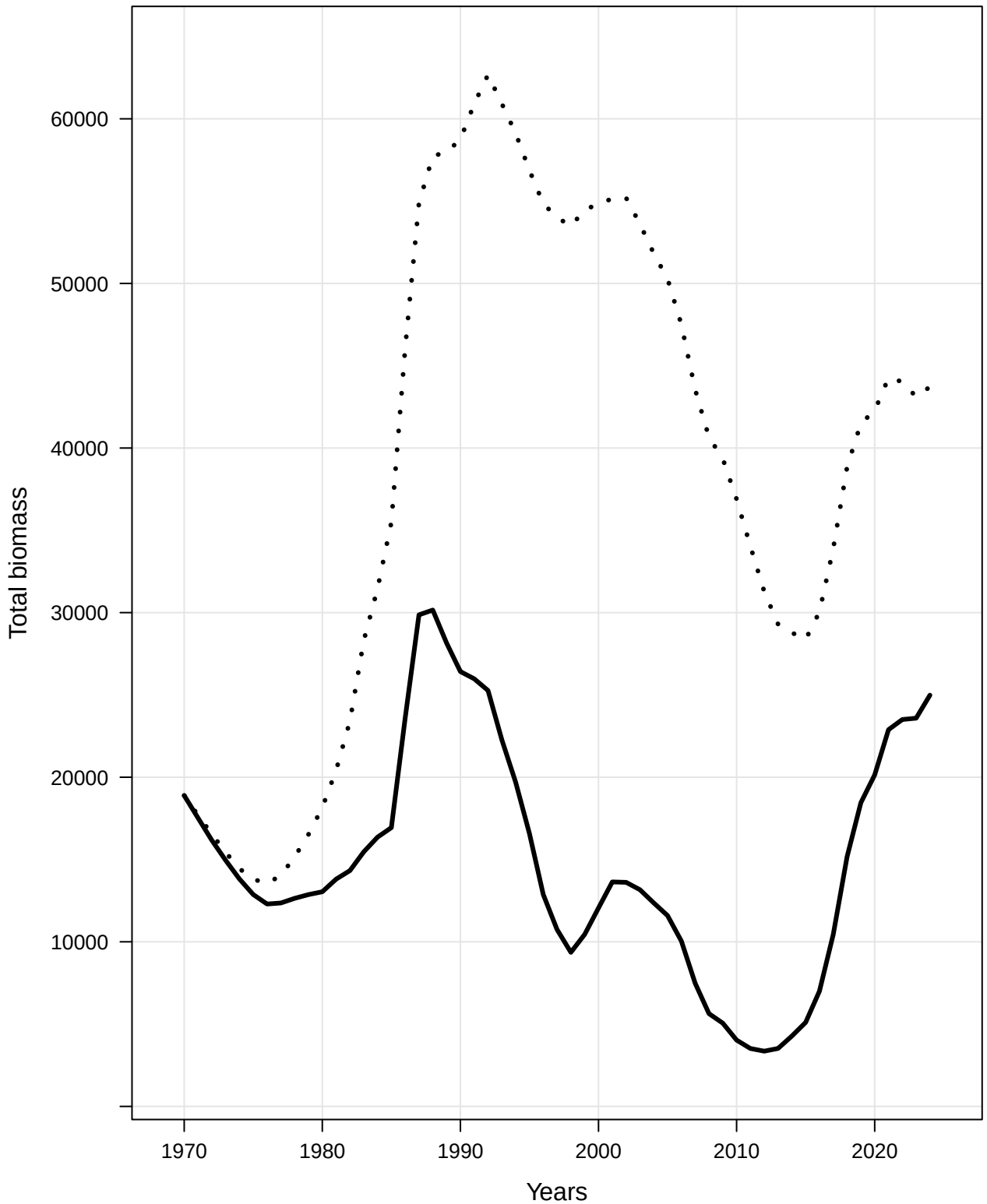


Stock Recruitment

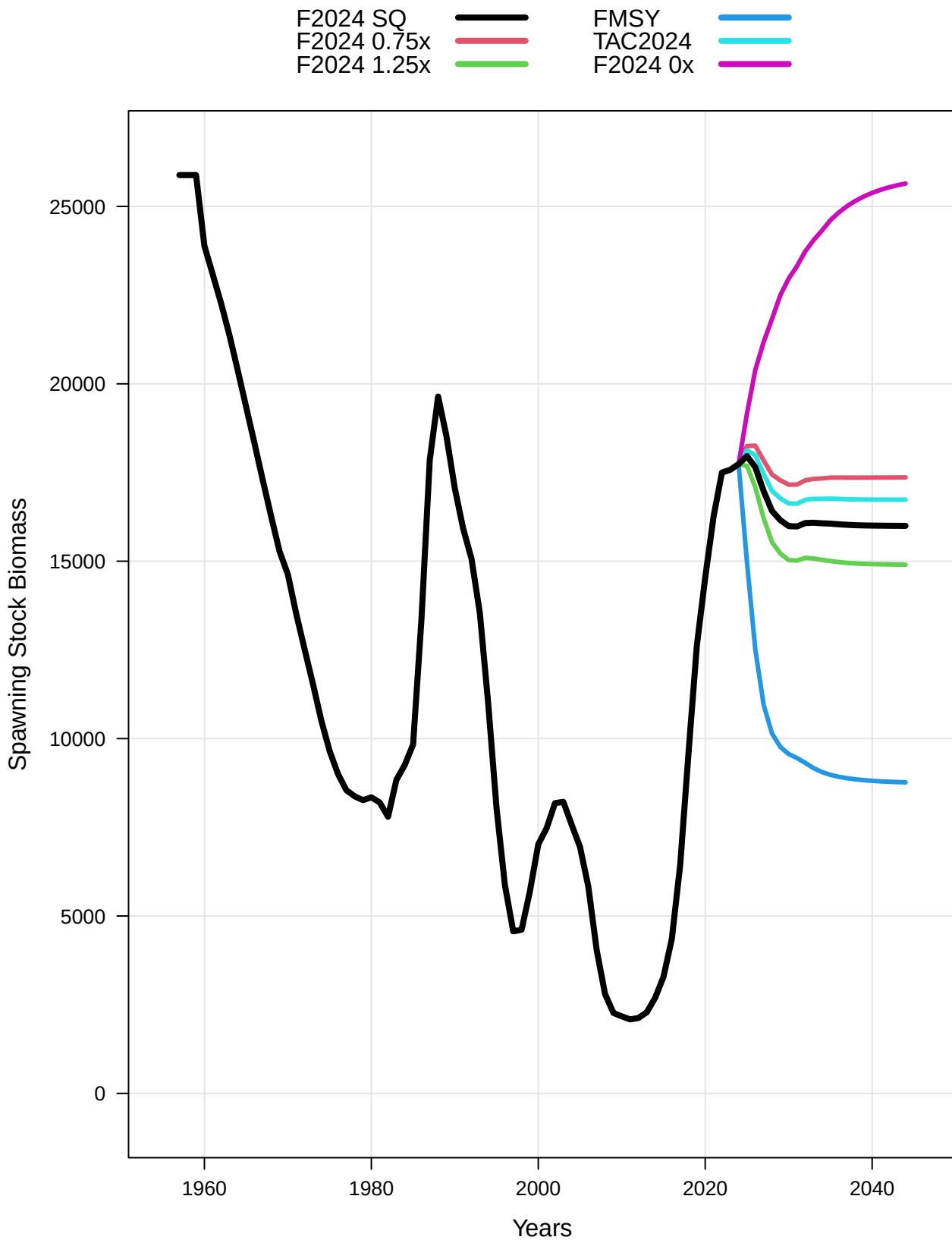


Comparing fished with unfished biomass

Fished — Unfished



SSB prediction



Catch prediction

F2024 SQ

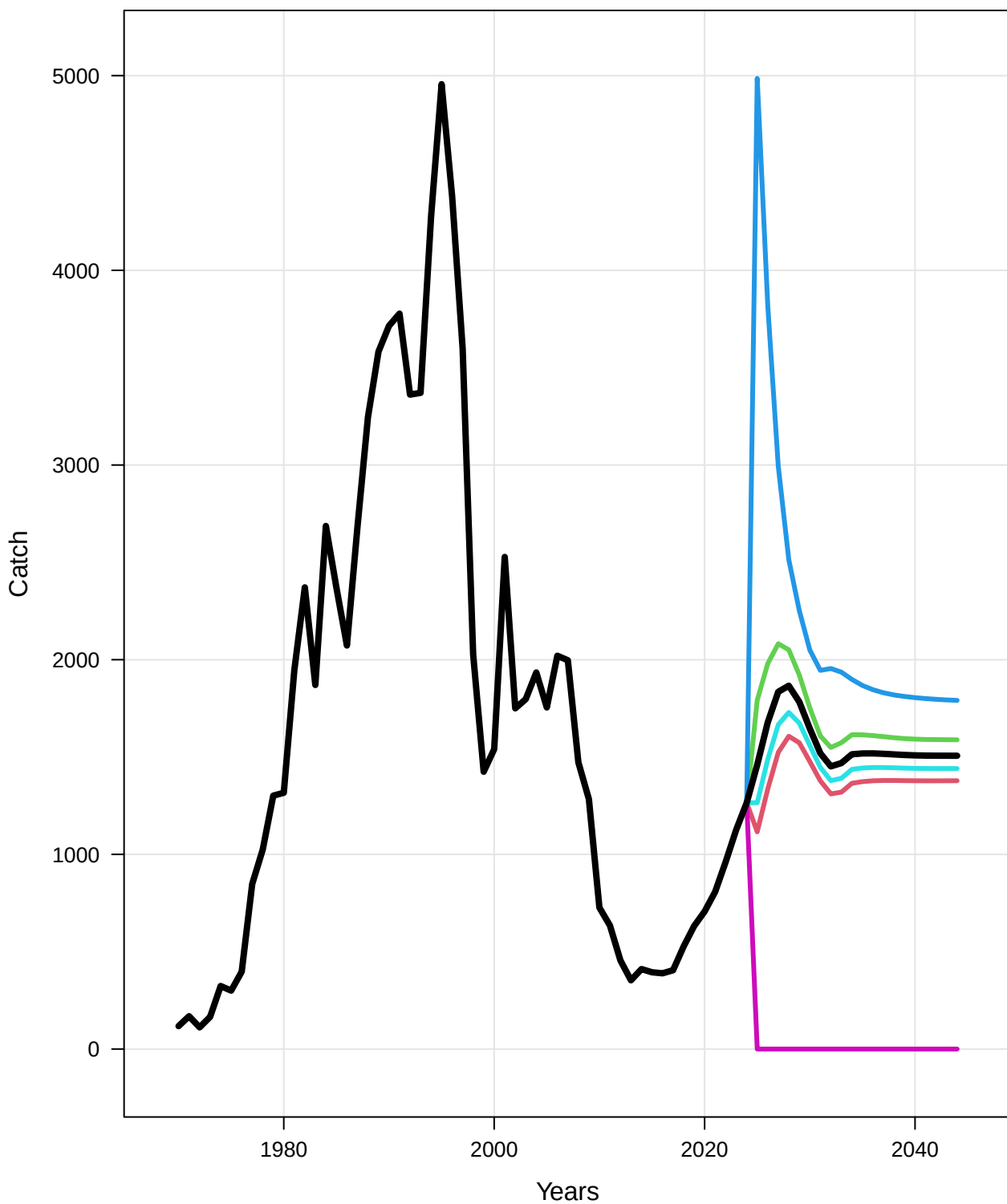
F2024 0.75x

F2024 1.25x

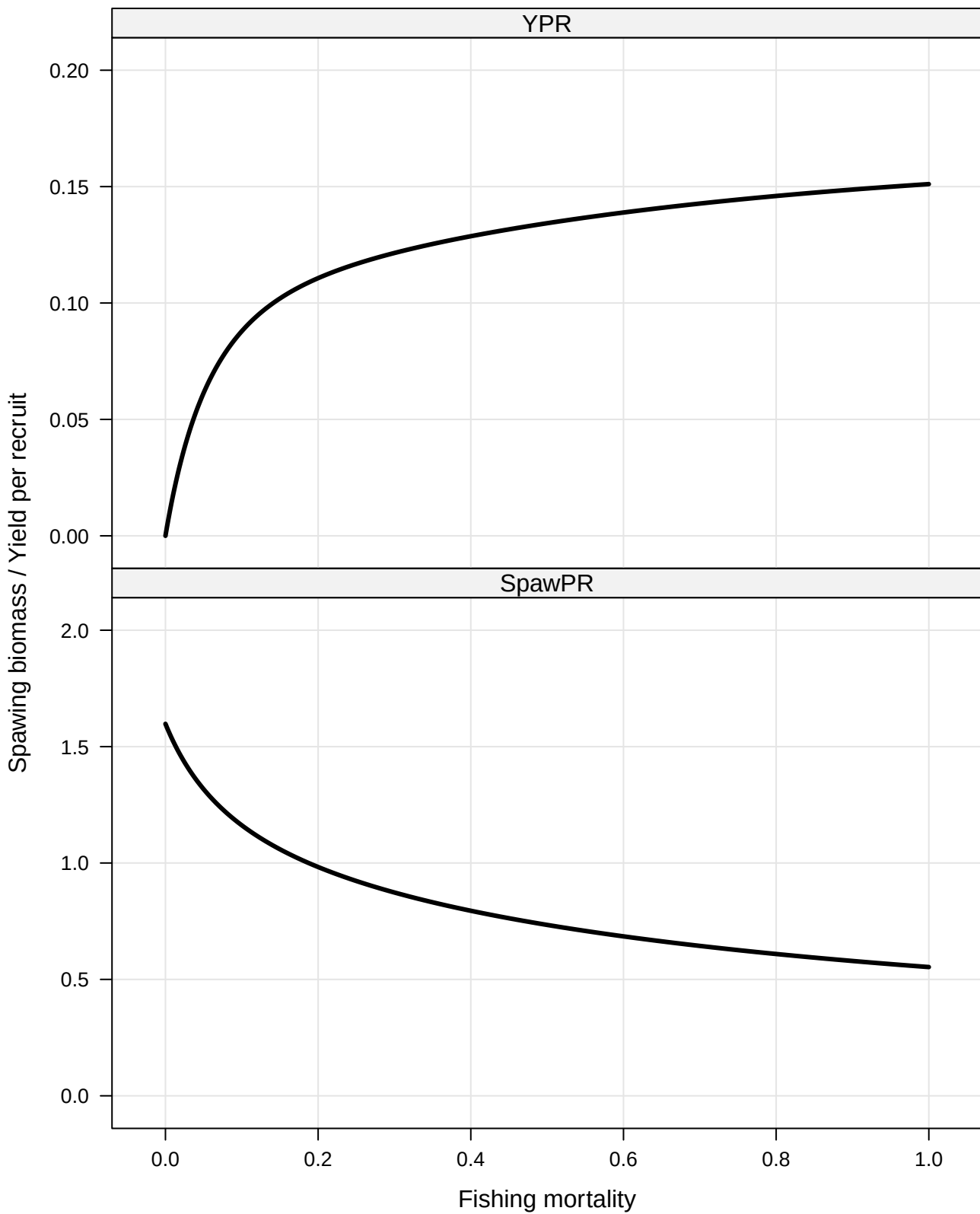
FMSY

TAC2024

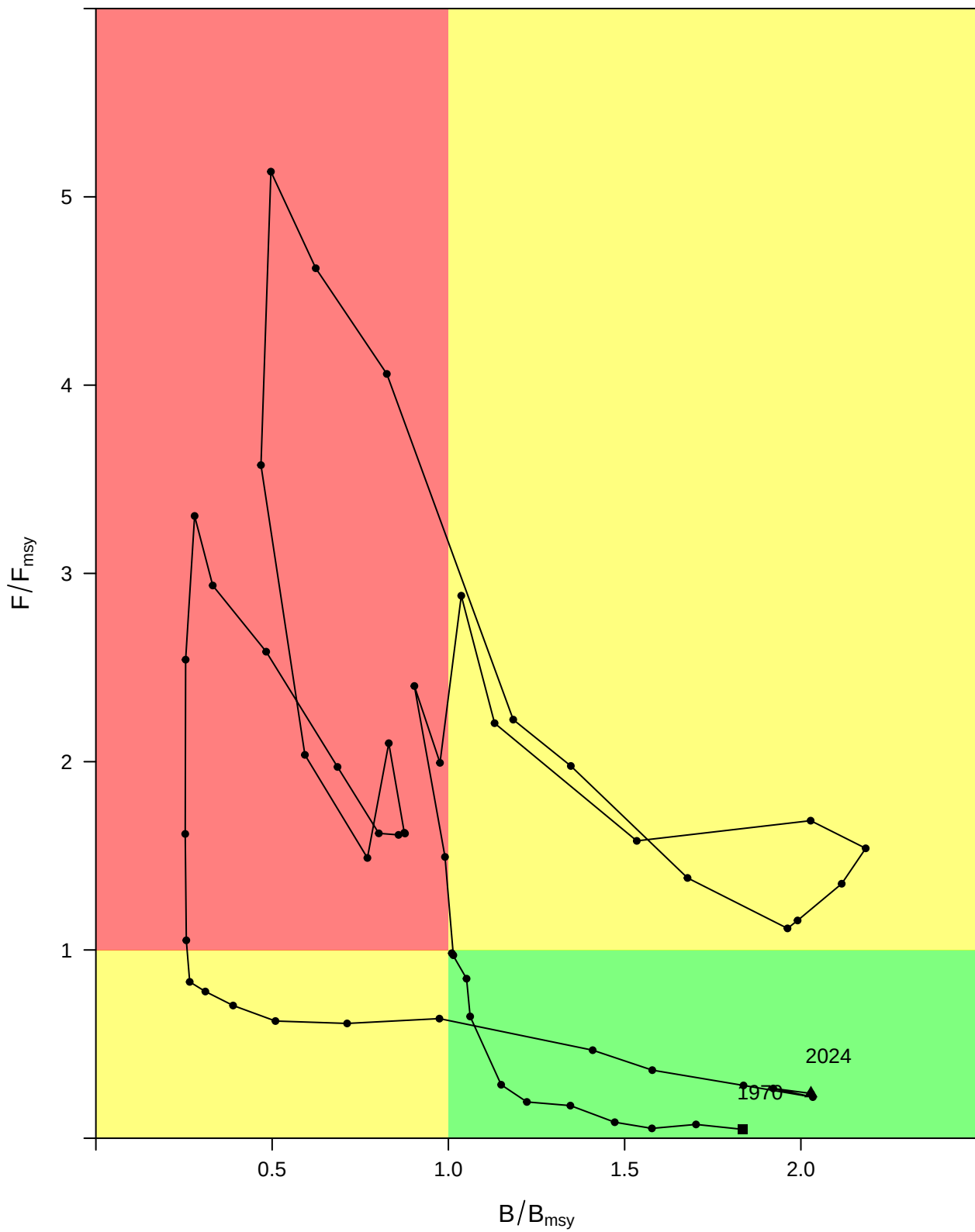
F2024 0x

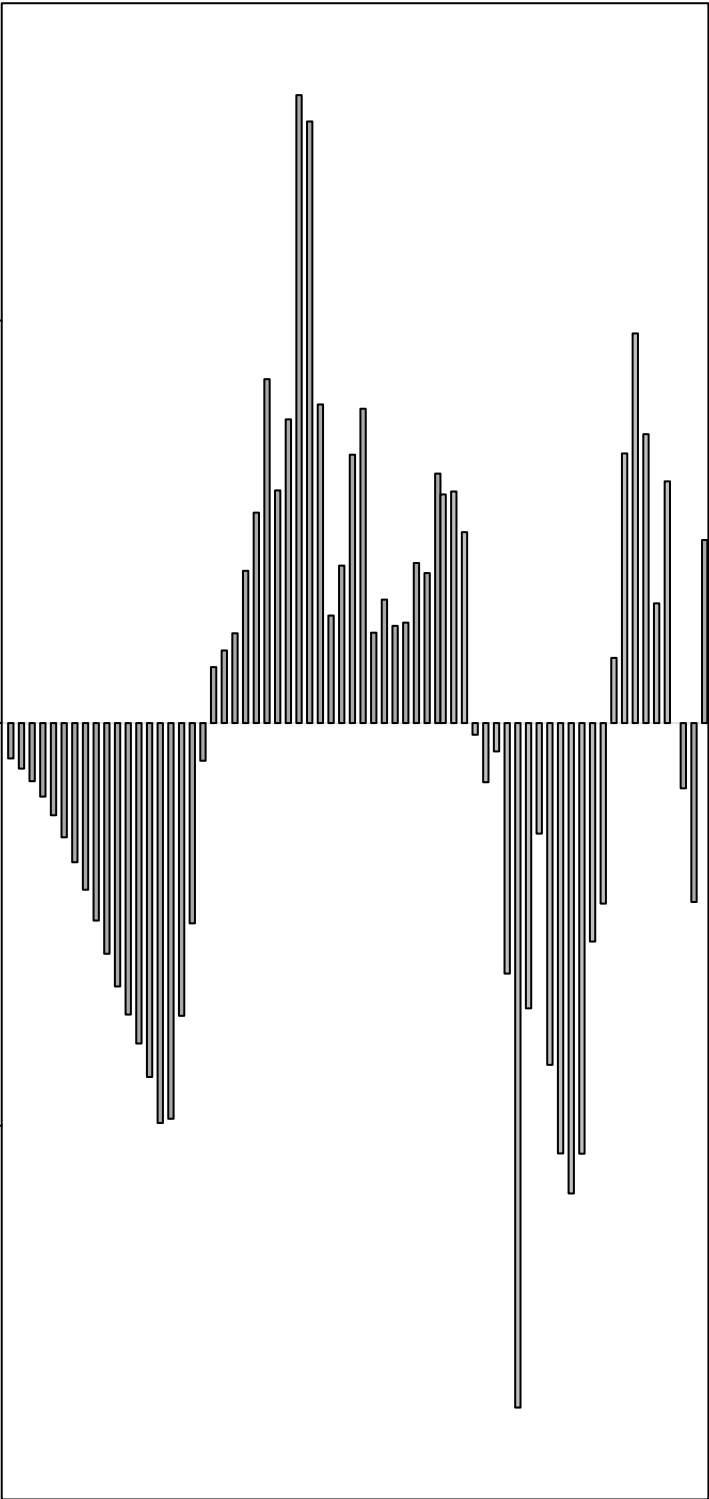


Yield and spawning stock biomass per recruit



Kobe plot





Simulated
2000 - 2021



